[B17 BS 1101]

I B. Tech I Semester (R 17) Regular Examinations

ENGLISH-I

(Common to all branches)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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UNIT-I

1. A) Correct the following sentences. (7M)

i. The machineries were expensive.
ii. Suppose, if you arrive late, you will miss the show.
iii. Choose the best of the two options.
iv. I enjoyed during the holidays.
v. I have seen him yesterday.
vi. The teacher gave us many advices.
vii.My dog is better than him.

B) a)Write appropriate quantifiers for each sentence. (3M)

i. The project is .... complicated than the last one
ii. I have to buy ....pairs of blue and black jeans soon.
iii. There is no ………. water in the bottle.

b) Re-write the sentences by using Gerunds, to-infinitives or plain infinitive forms. (4M)

i. I noticed him .... (write) picture postcards.
ii. I feel happy to be……… (sing) a song.
iii. They felt surprised to …….. (select) by the manager.
iv. …………(Garden) is a pleasant activity.

(OR)

2. A) Fill in the blanks using the appropriate forms of verbs given in the brackets. (7M)

i. In a fit of rage, she ____ up the letter. (Tear)
ii. We couldn't have _____ a better day for organizing the party. (Choose)
iii. It's high time you______ your mistake. (Realise)
iv. The poem 'The Gift of India'_____ (write) by Sarojini Naidu in 1915.
v. We ___ for five years now. (marry)
vi. When I _____ home, I found that there was no edible oil left. (go)
vii. The Journalist reported that the miscreants _____ a havoc in the city. (create)

B) a) Fill in the blanks by using appropriate conjunctions (3M)

i. Receptionists must be able to relay information ______ pass messages accurately.
ii. Mary is a member of the Historical Society ______ the Literary Society.
iii. My friend didn't work hard_____ he got through the exam.

b) Punctuate the following sentences. (4M)

i. sunil sharma is documentation development manager at cerner corporation one of the world's largest medical software developers
ii. As part of his job Sunil writes web-based content for Cerner.
iii. Hang him not leave him.
iv. my friend suresh who is in bengaluru has come today.

UNIT-II

3. A) Write one word substitutions to the following and write sentences by using them. Marks will be awarded only when both the points are correctly answered. (7M)

   i. Language which is confusing and unintelligible.
   ii. One who prepares plans for buildings.
   iii. A great lover of books
   iv. A person in charge of a museum
   v. A man who thinks only for himself
   vi. One who kills animals and sells their flesh
   vii. A person with a long experience in a specific field

B) a) Give synonyms for the following words and use them in your own sentences. (3M)

   i. Euphoria ii. Vicious iii. Ostentatious

   b) Give antonyms for the following words and use them in your own sentences. (4M)

   i. Truce ii. Terse iii. Supercilious iv. Emerge

(OR)

4. A) Give meanings for the following idioms and also use them in your own sentences. (7M)

   i. The cream of the crop ii. An arm and a leg
   iii. Hand in glove iv. Hue and cry
   v. Hard and fast vi. Explore all avenues
   vii. Spill the beans

B) a) Give synonyms for the following words and use them in your own sentences. (3M)

   i. Sacrilege ii. Pugnacious iii. Vitiate

   b) Give antonyms for the following words and use them in your own sentences. (4M)

   i. Succinct ii. stigmatize iii. recalcitrant iv. Adamant

UNIT-III

5. A) Read the following paragraph and answer the questions: (7M)

The study of history provides many benefits. First, we learn from the past. We may repeat mistakes, but, at least, we have the opportunity to avoid them. Second, history teaches us what questions to ask about the present. Contrary to some people's view, the study of history is not the memorization of names, dates, and places. It is the thoughtful examination of the forces that have shaped the courses of human life. We can examine events from the past and then draw inferences about current events. History teaches us about likely outcomes.

Another benefit of the study of history is the broad range of human experience which is covered. War and peace are certainly covered as are national and international affairs. However, matters of culture (art, literature, and music) are also included in historical study. Human nature is an important part of history: emotions like passion, greed, and insecurity have influenced the shaping of world affairs. Anyone who thinks that the study of history is boring has not really studied history.

   i. What is the central idea of this passage?
   ii. In the first paragraph, 'inferences' mean?
   iii. Which method of teaching history would the author of this passage support?
   iv. In the second paragraph, 'shaping of world affairs' Means.
   v. What is the conclusive thought of the author?
   vi. Give an appropriate title for the written discourse.
   vii. How reliable is the written history; and/or is it just 'his' story?

B) Develop a paragraph (200 words) based on the following hints and provide an appropriate title for the same. (7M)

As the 11th President of India---- the Indian National Congress------- 'people's president', he was--------. His contribution ---------Bharat Ratna. During ---------in India. He is the -------India: 2020 and Ignited Minds.

(OR)

6. A) Read the following paragraph and answer the questions: (7M)

Work expands so as to fill the time available for its completion. The general recognition of this fact is shown in the proverbial phrase, 'It is the busiest man who has time to spare.' Thus, an elderly lady at leisure can spend the entire day writing a postcard to her niece. An hour will be spent in finding the postcard, hunting for spectacles, half an hour to search for the address, an hour and a quarter in composition and twenty minutes in deciding whether or not take an umbrella when going to the pillar box in the street. The total effort that would occupy a busy man for three minutes, all told, may in this fashion leave another person completely exhausted after a day of doubt, anxiety and toil.

1. What happens when the time to be spent on some work increases?
2. Explain the sentence: 'Work expands so as to fill the time available for its completion.'
3. Who is the person likely to take more time to do work:
4. What is the total time spent by the elderly lady in writing a postcard?
5. What does the expression 'pillar box' stand for?
6. Suggest an appropriate title for the passage.
7. 'It is the busiest man who has time to spare' Elaborate the semantic content of it.

B) Develop a paragraph (200words) based on the following hints and provide an appropriate title for the same. (7M)

_____not luck but labor _________ Luck ___ever waiting_______ ;labour _____ strong-will turns up something. Luck _____ news of a legacy; labour _______ the foundation of competence. Luck _____on chance, labour ______ character.

UNIT-IV

7. A) Write an Essay on 'Terrorism, a social evil' (7M)

B) Draft an E-Mail to your friend about your career plans. (7M)

(OR)

8. A) Write an essay on 'Facing a book vis-à-vis Facebook' (7M)

B) Present an argument in about 250 words on 'Technology replacing Teachers'. Substantiate your argument with reasons. (7M)

UNIT-V

9. A) Write a feasibility report on 'Setting up a Water / Power Unit at your campus.' (7M)

ii. Write a report on Educational Tour

B) Draft a pamphlet on any Electronic home appliances/Places of tourists' interest/an Educational institution/ an exhibition. (7M)

(OR)

10.A) Write a feasibility report on 'Educational Tour'. (7M)

B) Write a letter to a renowned person, requesting him to be the Chief Guest for the cultural festival of your college. (7M)

[B17 BS 1101]

[B17 BS 1102]

I B. Tech I Semester(R17) Regular Examinations

MATHEMATICS-I

(Common to all branches)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
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UNIT – I

1. (a) Solve $\frac{dy}{dx} + (\tan x)\, y = (\sec x)\, y^3$. (7M+7M)
   (b) Find the orthogonal trajectories of the family of parabolas $ay^2 = x^3$.

(OR)

2. (a) Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$. (7M+7M)
   (b) A body originally at 80^0C cools down to 60^0C in 20 minutes, the temperature of air being 40^0C. What will be the temperature of the body after 40 minutes from the original?

UNIT - II

3. (a) Solve $(D^3 - D)y = 2x + 1 + 4\cos x$. (7M+7M)
   (b) $solve$ $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \log x$ by the method of variation of parameters.

(OR)

4. (a) $solve$ $(D^2 + 3D + 2)y = e^{e^x}$. (7M+7M)
   (b) The differential equation for a circuit in which self inductance and capacitance neutralize each other is $L\frac{d^2i}{dt^2} + \frac{i}{C} = 0$. Find the current i as a function of t , given that I is maximum current and i = 0 when t = 0.

UNIT - III

5. (a) Find $L\{t\cos at\}$ and $L\left\{ \int_0^t e^{-t}\cos t \; dt \right\}$. (7M+7M)
   (b) Using convolution theorem evaluate $L^{-1}\{ \frac{1}{(s+a)(s+b)}\}$.

(OR)

6. (a) Find $L^{-1}\{ \frac{5s+3}{(s-1)(s^2+2s+5)}\}$. (7M+7M)

   (b) Solve $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = e^{-t}$, $y(0) = y'(0) = 1$ by using Laplace transforms.

UNIT – IV

7. (a) If $U = \tan^{-1}\frac{x^3+y^3}{x-y}$ and $x U_X + y U_y = \sin 2U$, prove that

$x^2 U_{xx} + 2xy\, U_{xy} + y^2\, U_{yy} = 2\cos 3U \sin U.$ (7M+7M)

(b) If $u = x^2 - 2y^2$, $v = 2x^2 - y^2$ where $x = r\cos\theta$, $y = r\sin\theta$

show that $\frac{\partial(u,v)}{\partial(r,\theta)} = 6\, r^3 \sin 2\theta.$

(OR)

8. (a) Expand $x^2y + 3y - 2$ in powers of $(x-1)$ and $(y+2)$ using Taylor's theorem. (7M+7M)

(b) By using the method of differentiation under the integral sign prove that $\int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)}\, dx = \frac{\pi}{2}\log(1+a)$, $a \geq 0$.

UNIT – V

9. (a) Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$. (7M+7M)
(b) solve $(D^2 - DD' - 2D'^2)\, z = (y - 1)\, e^x$.

(OR)

10. (a) Solve $x(y - z)p + y(z - x)q = z(x - y)$. (7M+7M)
(b) solve $(D + D' - 1)(D + 2D' - 3)z = 3x + 6y + 4$.

[B17 BS 1102]

[B17 BS 1103]

I B. Tech I Semester (R 17) Regular Examinations

MATHEMATICS-II
(Mathematical Methods)

(Common to CSE,ECE & IT)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
*****

UNIT - I

1.a) Find a root of x^3-x-11=0 using the bisection method correct to three decimal places. (7M+7M)

b) Find the cube root of 41 using Newton-Raphson method.

(OR)

2. a) Find a real root of the equation x $\log_{10}$ x=1.2 by Regula-false method correct to three decimal places. (7M+7M)

b) Find the positive root of the equation 3x = cos x +1 by iteration method.

UNIT - II

3. a) Using Gauss forward difference formula, Find Y (8), from the following table (7M+7M)

X	0	5	10	15	20	25
Y	7	11	14	18	24	32

b) Find the interpolating polynomial f(x) for the data of the following table

x	0	1	4	5
f(x)	4	3	24	39

(OR)

4. a) Using Gauss backward formula, find f(42), from the following table (7M+7M)

X	20	25	30	35	40	45
f(x)	354	332	291	260	231	204

b) Using Lagrange's interpolation formula find Y(10) from the following table

x	5	6	9	11
Y	12	13	14	16

UNIT - III

5. a) Evaluate $\int_0^2 \frac{dx}{x^3+x+1}$ by using Simpsons 1/3rd rule with $h = 0.25$ (7M+7M)

b) Evaluate $y(0.8)$ using Runge Kutta method given $y' = (x+y)^{\frac{1}{2}}, y(0.4) = 0.41$

(OR)

6. a) A rocket is launched from the ground. Its acceleration a(t) measured every 5 seconds is tabulated below. Use trapezoidal rule to find the velocity and the position of the rocket at $t = 40$ seconds. (7M+7M)

t	0	5	10	15	20	25	30	35	40
a(t)	40.0	45.25	48.50	51.25	54.35	59.48	61.5	64.3	68.7

b) Given $y' = x + \sin y$, $y(0) = 1$, compute $y(0.2)$ and $y(0.4)$ with $h = 0.2$ using modified Euler's method.

UNIT – IV

7. a) Find the Fourier series to represent $f(x) = x - x^2 \; from \; x = -\pi \;\; to \;\; x = \pi.$ (7M+7M)

b) Obtain the sine series for $f(x) = x \;\; in \; 0 \le x \le \pi.$

(OR)

8. a) Obtain the Fourier series for the function $f(x) = \begin{cases} \pi x, & 0 \le x < 1 \\ \pi(2 - x), & 1 \le x \le 2 \end{cases}$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots\ldots\ldots = \frac{\pi^2}{8}$. (7M+7M)

b) Find half range cosine series for $f(x) = x(2 - x)$ in $0 < x < 2$.

UNIT – V

9. a) Find the Fourier Transform of $\frac{1}{\sqrt{|x|}}$. (7M+7M)

b) Find the Fourier integral representation for f(x)= $\begin{cases} 1 - x^2, & for|x| \le 1 \\ 0, & for|x| > 1 \end{cases}$

(OR)

10. a) Find the inverse Fourier transform $f(x)$ of $F_s(p) = \frac{p}{1+p^2}$. (7M+7M)

b) Find the Fourier cosine transform of e^{-ax}. Hence evaluate $\int_0^\infty \frac{\cos \lambda x}{x^2 + a^2} dx$

[B17 BS 1103]

[B17 BS 1104]

I B. Tech I Semester (R 17) Regular Examinations

ENGINEERING PHYSICS

(Common to CSE,ECE & IT)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT – I

1. (a) Describe the interference phenomena In thin transparent films for reflected light and obtain the conditions for maxima and minima. [7M]

   (b)Discuss the Fraunhoffer diffraction of monochromatic light at a single slit. [7M]

(OR)

2. (a)Describe, with a neat sketch, the design and working of Michelson's interferometer. [7M]

   (b)Explain how the resolving power of a grating can be determined. [7M]

UNIT – II

3. (a)Differentiate spontaneous and stimulated emission processes and obtain the Einstein's relation for spontaneous to stimulated emission coefficients. [7M]

   (b) Define numerical aperture of an optical fiber and derive an expression for the same. [7M]

(OR)

4. (a)With neat sketches, explain the principle and working of He – Ne gas laser system. [7M]

   (b)Explain the characteristics of lasers and mention the applications of lasers. [7M]

UNIT – III

5. (a)Discuss the electric fields induced due to time varying magnetic fields and deduce the Faraday's law. [7M]

   (b)Describe any one method of detecting ultrasonics and mention the important applications of Ultrasonics. [7M]

(OR)

6. (a) Explain the concept of displacement current, and describe the significance of Maxwell's equations. [7M]

   (b)What is magnetostriction and describe the magnetostriction method of producing Ultrasonics. [7M]

UNIT – IV

7. (a)What are matter waves and describe an experiment confirming the wave nature of electrons. [7M]

   (b)What are the salient features of Kronig - Penny model. [7M]

(OR)

8. (a)Explaning the physical significance of wave function of a particle derive the Schrodinger's time independent wave equation. [7M]

   (b)Using band theory of solids how do you classify the materials. [7M]

UNIT - V

9. (a)What is a unit cell and describe the different crystal systems possible in solids. [7M]
   (b)What are nano materials and explain the chemical vapour deposition method of fabricating nano materials. [7M]

(OR)

10. (a) Define packing fraction and deduce the packing fraction for a simple cubic structure. [7M]
    (b)Define the basic approaches of fabricating nano materials and discuss the sol-gel method. [7M]

[B17 BS 1104]

[B17 BS 1105]

I B. Tech I Semester (R 17) Regular Examinations

ENGINEERING CHEMISTRY

(Common to CIV, EEE & ME)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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UNIT - I

1. (a) Explain the mechanism of free radical Polymerization reaction with a suitable example. [7M]

(b)Distinguish between thermoplastic and thermosetting resins. [7M]

(OR)

2. (a)What are conducting Polymers? Discuss the applications of conducting Polymers. [7M]

(b)Write notes on Bu Na – S and Bu Na – N. [7M]

UNIT - II

3. (a) Explain the Proximate analysis of coal and give its significance. [7M]

(b) Explain the fractional distillation of crude oil. [7M]

(OR)

4. (a)Write notes on (i) Knocking (ii) CNG [7M]

(b)How Synthetic Petrol can be prepared by Bergius Process**.** [7M]

UNIT - III

5. (a) Explain the mechanism of electrochemical theory of corrosion with neat diagram. [7M]

(b) Describe briefly about cathodic Protection. [7M]

(OR)

6. (a) Explain Hydrogen – Oxygen fuel cell with neat cell diagram [7M]

(b)Discuss on various constituents of Paint. [7M]

UNIT - IV

7. (a) What is hardness? How it is determined by EDTA method? Explain. [7M]

(b) Describe with equations how water can be softened using Lime & Soda Process [7M]

(OR)

8. (a) Discuss various sterilizing methods used in municipal water treatment. [7M]

(b) Illustrate the reverse osmosis Process with a neat diagram. [7M]

UNIT - V

9. (a) Discuss chemistry involved in setting and hardening of cement? [7M]

(b) What are refractories? Discuss the classification of refractories . [7M]

(OR)

10. (a) Write the engineering applications of Liquid Crystals . [7M]

(b) Explain the stoichiometric defects in crystals. [7M]

[B17 BS 1105]

[B17 CS 1101]

I B. Tech I Semester (R 17) Regular Examinations

COMPUTER PROGRAMMING USING C

(Common to CSE,ECE & IT)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.

****

UNIT-I

1. a) Discuss about computer languages. [7M]
   b) Explain c tokens. [7M]

(OR)

2. a) Explain different c operators. [7M]
   b) write about algorithm, pseudo code and flowchart. [7M]

UNIT-II

3. a) Discuss various looping techniques in c. [7M]
   b) Write a c program for summation of n numbers. [7M]

(OR)

4. a) Explain 2-D arrays and character arrays in c. [7M]
   b) Write a c program to find frequency of characters of a string. [7M]

UNIT-III

5. a) Explain parameter passing techniques in c. [7M]
   b) Write a c program for towers of Hanoi using recursive function. [7M]

(OR)

6. a) Explain storage classes in c. [7M]
   b) Write a c program for Fibonacci series using recursive function. [7M]

UNIT-IV

7. a) What is a pointer? How pointer variables are initialized. [7M]
   b) Write a program to print command line arguments on the screen. [7M]

(OR)

8. a) Discuss character pointers with examples. [7M]
   b) Write a c program to pass pointer variables as function arguments. [7M]

UNIT-V

9. a) Explain the difference between structure and union and write a program to find sum of marks in 3 subjects for a student using structures. [7M]
   b) Explain different bit-wise operators in c. [7M]

(OR)

10. a) Explain about the input and output operations of a file. [7M]
    b) Write a c program to open a file and to print its contents on screen. [7M]

[B17 CS 1101]

[B17 CE 1101]

I B. Tech I Semester (R 17) Regular Examinations

ENVIRONMENTAL STUDIES

(Common to all Branches)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
*****

UNIT-I

1 a).	Define Environment. Write the scope and importance of the environmental studies.	[7M]
b).	Elucidate the concept of Global Environmental crisis.	[7M]
	(OR)	
2 a).	What is an ecosystem? Write the structure and functions of an ecosystem.	[7M]
b).	Write a brief note on forest resources.	[7M]
	UNIT-II	
3 a).	Describe the values of Biodiversity.	[7M]
b).	Write about in-situ and ex-situ conservation.	[7M]
	(OR)	
4 a).	Describe Biogeographical Classification of India.	[7M]
b).	India as a mega-diversity habitat – Explain	[7M]
	UNIT-III	
5 a).	Effects of modern agriculture on land.	[7M]
b).	What are the benefits and problems of dams?	[7M]
	(OR)	
6 a).	Write about floods and droughts?	[7M]
b).	Discuss the impact of energy use on environment.	[7M]
	UNIT-IV	
7 a).	What are the causes, effects and control measures of air pollution?	[7M]
b).	What is solid waste management? Explain its methods.	[7M]
	(OR)	
8 a).	Elucidate the results of population growth on environment?	[7M]
b).	Write notes on Rain water harvesting with a neat sketch	[7M]
	UNIT-V	
9 a).	Mention the different environmental acts and write about one.	[7M]
b).	Write notes on Environmental impact Assessment.	[7M]
	(OR)	
10 a).	Write short notes on any two environmental case studies.	[7M]
b).	Write a report on a visit to an environmental polluted area?	[7M]

[B17 CE 1101]

[B17 ME 1101]

I B. Tech I Semester (R 17)Regular Examinations

ENGINEERING MECHANICS

(Common to CIV,EEE & ME)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
Assume the missing data if any, suitably

*****

UNIT-1

1. (a) State and prove Varignon's theorem. [7 M]
(b) Two cylinders of diameter 100 mm and 50 mm, weighing 200 N and 50 N, respectively are placed in a trough as shown in Figure 1. Assuming smooth surfaces, find the reactions at the points of supports 1, 2, 3 and 4. [7 M]

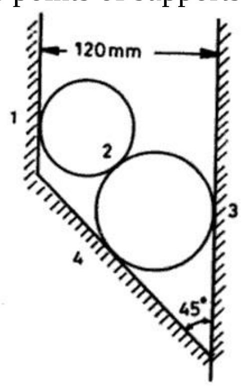


Figure 1

[OR]

2 (a) A string ABC of length *l* carries a small pulley C from which a Load W is suspended as shown in Figure 2. The string hangs between two vertical walls which are at a distance *d* apart. The end A is higher than the end B by height *h*. Find the position of equilibrium defined by the angle α. Assume *d*= *l*/2 and *h*=*l*/4. [7 M]

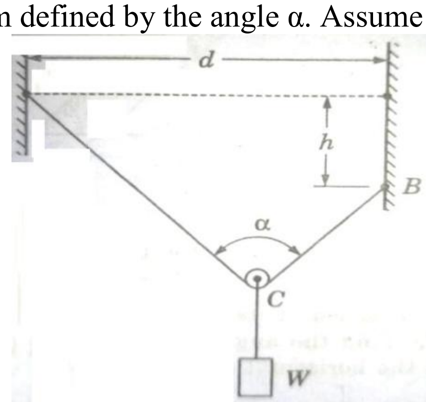


Figure 2

(b) Two identical prismatic bars AB & CD each weighing 200 N are welded together to form a Tee and are suspended in a vertical plane as shown in Figure 3. Calculate the values of the θ that the bar AB will make with the vertical when a vertical load of 200 N is applied at D. [7 M]

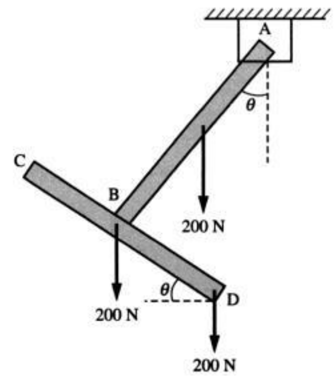


Figure 3

UNIT-II

3 (a) Derive the centriod of a wire bend in the form of a sector of an arc by taking the radius as ‘r’ and angle of sector as ‘θ’. [7 M]

(b) Determine the centriod of the shaded segment for Figure 4 by taking a = 18 m and $\alpha = 45^0$. [7 M]

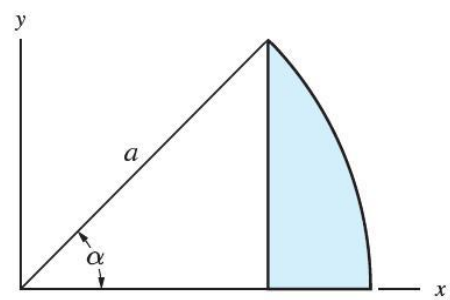


Figure 4

[OR]

4 (a) Derive the moment of inertia of triangle about its centriodal axis and also deduce the same about its base. [7 M]

(b) Determine the moment of Inertia of the T-section shown in Figure 5 about its centroidal axis. [7 M]

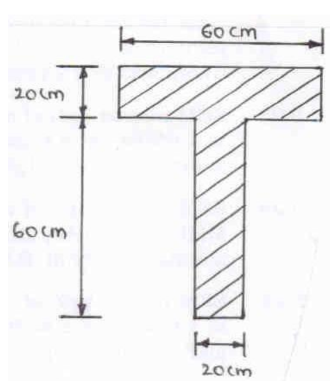


Figure 5

UNIT-III

5 (a) Explain the terms angle of repose, cone of friction and write the laws of friction. [7 M]

(b) Referring to the Figure 6 given above, determine the least values of the force P to cause motion to impend right wards. Assume the coefficient of friction under the blocks to be 0.2 and pulley to be frictionless. [7 M]

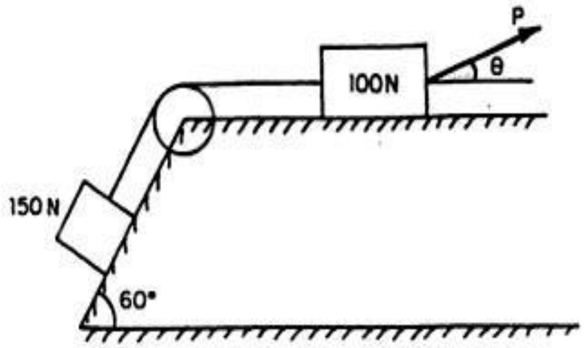


Figure 6

[OR]

6 (a) A uniform ladder 5m long on a horizontal ground and leans against a smooth vertical wall at an angle of 70^0 with the horizontal. The weight of the ladder is 90 N and acts at its middle. The ladder is at the point of sliding, when a man weighing 75N stands on a rung 3.5m from the top of the ladder. Calculate the co-efficient of the friction between the ladder and the floor. [7 M]

(b) Find out the forces in all the members of a pin jointed truss as shown in Figure 7 by using method of Joints. [7 M]

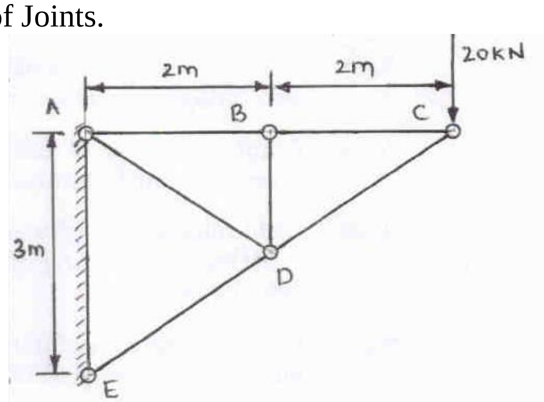


Figure 7

UNIT-IV

7 (a) A stone is dropped from the top of a tower 60 m high. At the same instant, another stone is thrown vertically upwards from the foot of tower to meet the first stone at a height of 18 m. Determine (i) the time when the two stones meet; (ii) the velocity with which the second stone was thrown up. [7 M]

(b) Weight W and 2W are supported in a vertical plane by a string and pulleys arranged as shown in Figure 8. Find the magnitude of an additional weight Q applied on the left which will give a downward acceleration a = 0.1g to the weight W. [7 M]

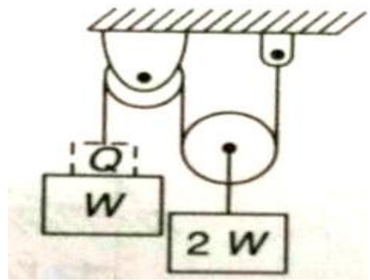


Figure 8

[OR]

8 (a) Define Time of Flight, Range and Maximum Height of a projectile. [7 M]

(b) Derive the general equation of projectile motion. [7 M]

UNIT-V

9 (a) A flywheel is rotating at 150 R.P.M. and after 8 seconds it is rotating at 120 R.P.M.. If the retardation is uniform, determine number of revolutions made by the flywheel and the time taken by the flywheel before it comes to rest from the speed of 150 R.P.M. [7 M]

(b) A rotor of weight W = 1720 N and radius of gyration k = 100 mm is mounted on a horizontal shaft and set in rotation by a falling weight W = 1720 N as shown in Figure 9. If the system is released from rest, find the velocity of the block after it has fallen through a distance of 3 m. [7 M]

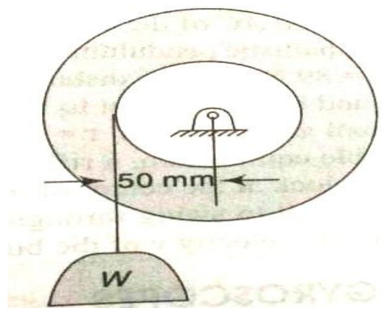


Figure 9

[OR]

10 (a) A body is rotating with an angular velocity of 8 radian/s. After 5 seconds, the angular velocity of the body becomes 28 radian/s. determine the angular acceleration of the body. [7 M]

(b) Three bodies, a sphere, a cylinder and a hoop each having the same mass and radius are released from rest from an inclined plane of angle θ. Determine the velocity of each of the bodies after it has rolled down the incline plane through a distance s. [7 M]

[B17 ME 1101]

[B17 ME 1102]

I B. Tech I Semester (R 17) Regular Examinations

ENGINEERING DRAWING

(Common to CIV,EEE & ME)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

Assume the missing data if any, suitably

*****

UNIT-I

1. An inelastic string 145 mm long has its one end attached to the circumference of a circular disc of 40 mm diameter. Draw the curve traced out by the other end of the string, when it is completely wound around the disc, keeping the string always tight. [14 M]

[OR]

2. Two fixed points A and B are 100mm apart, Trace the complete path of a point P moving (in the same plane as that of A and B) in such a way that the sum of its distance from A and B is always the same and equal to 125mm. Name the curve and draw another curve parallel to and 25mm away from this curve. [14 M]

UNIT-II

3. A line AB, of 80 mm long has its end **A**, 15 mm in front of VP and 20 mm above HP. The other end B is 40 mm above HP and 50 mm in front of VP. Draw the projections of the line and determine the inclinations of the line with HP and VP. [14 M]

[OR]

4. (a) Draw the projections of a 75mm long straight line in the following positions: (i) parallel to and 30mm above the HP and in the VP; (ii) perpendicular to the VP, 25mm above the HP and its one end in the VP; (iii) Inclined at 30^0 to the HP and its one end 20mm above it, parallel to and 30mm in front of the VP. [7 M]

(b) Draw the projections of the following points on the same ground line, keeping the projectors 25mm apart. (i) Point A in the HP and lying 20mm behind the VP; (ii) Point B is 40mm above the HP and 25mm in front of the VP; (iii) Point C is 25mm below the HP and 25mm behind the VP; (iv) Point D is 15mm above the HP and 50mm behind the VP. [7 M]

UNIT-III

5. Draw a rhombus of diagonals 100 mm and 60 mm long, with the longer diagonal horizontal. The figure is the top view of a square of 100mm long diagonals, with a corner on the ground. Draw its front view and determine the angle which its surface makes with the ground. [14 M]

[OR]

6. A semicircular plate of 40mm diameter has its straight edge in the VP and inclined at 45^o to the HP, the surface of the plate makes an angle of 30^o with the VP. Draw its projections. [14 M]

UNIT-IV

7. A hexagonal pyramid, base 25mm side and axis 50mm long, has an edge of its base on the ground. Its axis is inclined at 30^0 to the ground and parallel to the VP. Draw its projections. [14 M]

[OR]

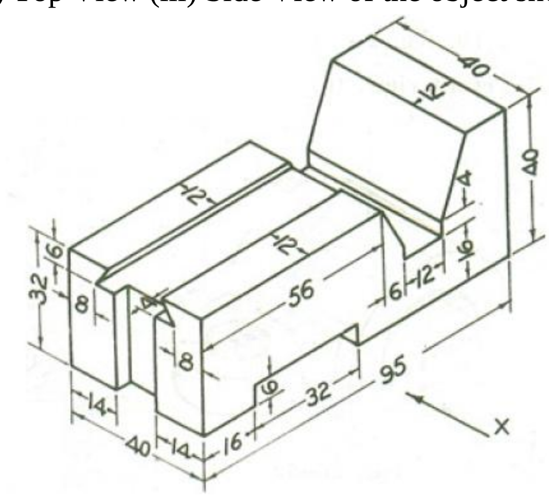
8. Draw the projections of a cylinder 75mm diameter and 100mm long, lying on the ground with its axis inclined at 30^o to the VP and parallel to the ground. [14 M]

UNIT-V

9. A square pyramid with base side 40mm and height 60mm is resting on a cube of sides 50mm, the axes of the cube and the pyramid being in the same line. Two sides of the base of the pyramid are parallel to the edges of the cube. Draw the isometric view. [14 M]

[OR]

10. Draw (i) Front View (ii) Top View (iii) Side View of the object shown below: [14 M]



All the dimensions are in mm

[B17 BS 1201]

I B. Tech II Semester (R 17) Regular Examinations

ENGLISH-II

(Common to all Branches)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.

UNIT-I

1. A. i) Describe how education is the greatest resource. (4M)
ii) Write a brief note on the great contribution made by Kalam to the science and technology. (3M)
B. Imagine that you are a builder and draft a letter of tender quotation to the Chief Engineer of Department of Roads and Buildings of Karnataka for constructing an administrative building. (7M)

(OR)

2. A. i) What, according to the author, is the source of problems for civilizations? (4M)
ii) Who had most influenced the value system of Kalam when he was young? (3M)
B. Imagine that you are the managing Director of a big company that manufactures electronic goods like Music systems, DVDs, LCDs. Write a business letter addressed to the Board of directors requesting them to attend a meeting to be held in the ensuing month. Give the agenda of the meeting also. (7M)

UNIT-II

3. A. i) What is the layman's view of atomic bomb? How right is he in thinking so? Who do you think is to be held responsible for the destruction created by technology? Support your opinion with suitable examples. (4M)
ii)What were some of the changes that Raman had initiated at the Indian Institute of Science? (3M)

B. Make notes on the following passage. (7M)
Here is an excerpt form one of Abdul Kalam's essays.
Knowledge has many forms and it is available at many places. It is acquired through education, information, intelligence and experience. It is available in academic institutions with teachers, in libraries, in research papers, seminar proceedings and in various organizations and work places with workers, managers, in drawings, in process sheets and on the shop floors. Knowledge, though closely linked to education, comes equally from learning skills, such as those possessed by our artists, craftsmen, hakims, vaidyas, philosophers and saints, as also our housewives. Knowledge plays a very important role in their performance and output too. Our heritage and history, the rituals, epics and traditions that form part of our consciousness are also vast resources of knowledge as are our libraries and universities. There is an abundance of unorthodox, earthy wisdom in our villages. There are hidden treasures of knowledge in our environment, in the oceans, bio-reserves and deserts, in the plant and animal life. Every state in a country has a unique core competence for a knowledge society

(OR)

4. A. i) Describe any modern invention with its positive and negative effects on the society. (4M)
ii)List out the awards and achievements of Sir C.V. Raman. (3M)
B. Make notes on the following passage. (7M)
It is not luck but labor that makes a man. Luck, says an American writer, is ever waiting for something to turn up; labour with keen eyes and strong will always turns up something. Luck lies in bed and wishes the postman would bring him news of a legacy; labour turns out at six and with busy pen and ringing hammer lays the foundation of competence. Luck whines, labour watches. Luck relies on

chance, labour on character. Luck slips downwards to self-indulgence; labour strides upwards and aspires to independence. The conviction, therefore, is extending that diligence is the mother of good luck. In other words, that a man's success in life will be proportionate to his efforts, to his industry, to his attention to small things.

UNIT-III

5. A.(i)How should one avoid culture shock before experiencing it when one goes to a new place? What precautions would help in living peacefully in a new place of new culture? (4M)
(ii)Explain in brief Baba's theory on the hitting of cosmic rays on earth's atmosphere (3M)

B. Write a paragraph on one of the following ideas. (7M)
i) Facebook ii) Barking dog seldom bites.

(OR)

6.A i) How does a person become a cultural entity ? (4M)
ii) Imagine that you have been elected as the Cultural Secretary of the Students' Association and you have to give a ten-minute speech outlining your plans for the academic year. Write out your speech in about 75 words. (3M)
B. Write an essay on Homy Baba's life and his academic and professional journey. (7M)

UNIT-IV

7. A i) How does Shirley Jackson trivialize the grave practice of the communities traditional stoning and what message might Jackson be trying to convey to the reader through the treatment of the characters' behavior? (4M)
ii) What were two types of services devised by the British in the Indian Education Services? Why? (3M)

B. Rewrite the following sentences correcting the errors: (7M)
i. He plays football when he was free
ii. He drunk coffee everyday when he was young
iii. Had your breakfast in the morning?
iv. He drunk coffee everyday when he was young.
v. Had your breakfast in the morning?
vi. Why haven't you been along with me for the event last month?
vii. Never I have seen such a person!

(OR)

8. A. i) What is black box? Who made it? When and why is it significant? (4M)
ii) Fill in the blanks with appropriate prepositions. (3M)
a. She was senior___ me when we were with the academic project ______some time.
b. One who believes __ and a devotee __ God is a theist.
c. He is angry__ her behavior as she always lies ___ him.
d. Write an essay on the contribution of J.C. Bose to the field of science. (7M)

UNIT-V

9. A. i) How did the relationship between Microsoft and IBM begin? (4M)
ii) Collocate the given words of the list A with those of the list B.

A	B	
i.Resounding	enemies	( )
ii. Bitter	success	( )
iii. Death	blow	( )

B. Write an essay on Dr.Prapulla Chandra Ray's life and his academic and professional journey (7M)

(OR)

10. A i) Describe How Gates worked for the development of Microsoft. (4M)
ii)Describe the compound Ray discovered. What are its properties? (3M)
B. Write a business report on ‘Setting up a Pharmaceutical Lab and Manufacturing Unit at Visakhapatnam, Andhra Pradesh. (7M)

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[B17 BS 1202]

I B. Tech II Semester (R 17) Regular Examinations

MATHEMATICS-II
(Mathematical Methods)

(Common to CIV, EEE & ME)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT - I

1.a) Find a root of x^3-x-11=0 using the bisection method correct to three decimal places. (7M+7M)

b) Find the cube root of 41 using Newton-Raphson method.

(OR)

2. a) Find a real root of the equation x $\log_{10}$ x=1.2 by Regula-false method correct to three decimal places. (7M+7M)

b) Find the positive root of the equation 3x = cos x +1 by iteration method.

UNIT - II

3. a) Using Gauss forward difference formula, Find Y (8), from the following table (7M+7M)

X	0	5	10	15	20	25
Y	7	11	14	18	24	32

b) Find the interpolating polynomial f(x) for the data of the following table

x	0	1	4	5
f(x)	4	3	24	39

(OR)

4. a) Using Gauss backward formula, find f(42), from the following table (7M+7M)

X	20	25	30	35	40	45
f(x)	354	332	291	260	231	204

b) Using Lagrange's interpolation formula find Y(10) from the following table

x	5	6	9	11
Y	12	13	14	16

UNIT - III

5. a) Evaluate $\int_0^2 \frac{dx}{x^3+x+1}$ by using Simpsons 1/3rd rule with $h = 0.25$ (7M+7M)

b) Evaluate $y(0.8)$ using Runge Kutta method given $y' = (x+y)^{\frac{1}{2}}, y(0.4) = 0.41$

(OR)

6. a) A rocket is launched from the ground. Its acceleration a(t) measured every 5 seconds is tabulated below. Use trapezoidal rule to find the velocity and the position of the rocket at t = 40 seconds. (7M+7M)

t	0	5	10	15	20	25	30	35	40
a(t)	40.0	45.25	48.50	51.25	54.35	59.48	61.5	64.3	68.7

b) Given y' = x + sin y, y(0) = 1, compute y(0.2) and y(0.4) with h = 0.2 using modified Euler's method.

UNIT – IV

7. a) Find the Fourier series to represent $f(x) = x - x^2 \; from \; x = -\pi \;\; to \;\; x = \pi.$ (7M+7M)

b) Obtain the sine series for $f(x) = x \;\; in \; 0 \le x \le \pi.$

(OR)

8. a) Obtain the Fourier series for the function $f(x) = \begin{cases} \pi x, & 0 \le x < 1 \\ \pi(2-x), & 1 \le x \le 2 \end{cases}$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots\ldots\ldots = \frac{\pi^2}{8}.$ (7M+7M)

b) Find half range cosine series for $f(x) = x(2-x)$ in $0 < x < 2.$

UNIT – V

9. a) Find the Fourier Transform of $\frac{1}{\sqrt{|x|}}.$ (7M+7M)

b) Find the Fourier integral representation for f(x)= $\begin{cases} 1 - x^2, \; for |x| \le 1 \\ 0, \; for |x| > 1 \end{cases}$

(OR)

10. a) Find the inverse Fourier transform $f(x)$ of $F_s(p) = \frac{p}{1+p^2}.$ (7M+7M)

b) Find the Fourier cosine transform of e^{-ax}. Hence evaluate $\int_0^\infty \frac{\cos \lambda x}{x^2 + a^2} dx$

[B17 BS 1202]

[B17 BS 1203]

I B. Tech II Semester (R 17) Regular Examinations

MATHEMATICS-III

(Common to all Branches)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

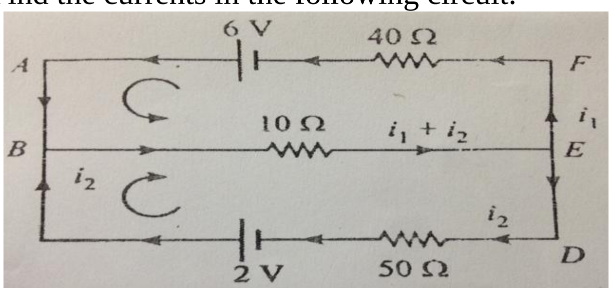
All questions carry equal marks.

*****

UNIT - I

1.a) Solve the system of equations $20x + y - 2z = 17, 3x + 20y - z = -18,$ $2x - 3y + 20z = 25$ by Gauss –Siedel method. (7M+7M)

b) Find the currents in the following circuit.



(OR)

2. a) Solve the system of equations 10x + y+z=12, 2x+10y+z=13, 2x+2y+10z=14 by Gauss-elimination method. (7M+7M)

b) Define rank and find the rank of the matrix A by reducing it in to its normal form where

A is: $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$.

UNIT - II

3. a) Verify Cayley-Hamilton theorem and find the inverse of the matrix

$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$. (7M+7M)

b) Reduce the quadratic form $2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ to canonical form by orthogonal transformation and hence find rank, index, signature and nature of the quadratic form.

(OR)

4. a) Find the eigen values and the corresponding eigen vectors of the matrix

$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$. (7M+7M)

b) If A = $\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, use Cayley-Hamilton theorem to find the value of $2A^5 - 3A^4 + A^2 - 4I$. Also find the inverse of A.

UNIT - III

5. a) Evaluate $\int_0^a \int_{\frac{x^2}{a}}^{2a-x} xy^2 \ dydx$ by changing the order of integration. (7M+7M)

b) Establish the relation between Beta and Gamma functions.

(OR)

6. a) Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx\, dy$ by changing in to polar coordinates and hence deduce $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$. (7M+7M)

b) Express $\int_0^1 x^m (1-x^n)^p \, dx$ in terms of Γ functions and hence evaluate

$\int_0^1 x^5 (1-x^3)^{10} \, dx.$

UNIT – IV

7. a) Find the directional derivative of $\emptyset(x,y,z) = x^2yz + 4xz^2$ at the point (1,-2,-1) in the direction of the normal to the surface $f(x,y,z) = x \log z - y^2$ at (-1,2,1). (7M+7M)

b) Prove that div(grad r^n) = n (n+1) r^{n-2} and curl(grad $\emptyset$) = 0 for any scalar function $\emptyset$.

(OR)

8. a) Show that the vector field $\bar{F} = (x^2 + xy^2)\bar{I} + (y^2 + x^2y)\bar{J}$ is conservative and find the scalar potential function. (7M+7M)

b) Find the constants a, b such that the surfaces $5x^2 - 2yz - 9x = 0$ and $ax^2y + bz^3 = 4$ cut orthogonally at(1,-1,2).

UNIT – V

9. a) Evaluate by Green's theorem $\oint_C [(y - \sin x)dx + \cos x \, dy]$ where C is the triangle enclosed by the lines y=0, x=π /2, y=2x / π. (7M+7M)

b) State Gauss Divergence theorem and use it to evaluate $\iint_S \bar{u}.\bar{n} \, ds$
where $\bar{u} = \bar{r} = x\bar{I} + y\bar{J} + z\bar{K}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = 9$.

(OR)

10. a) State Green's theorem in a plane and apply the theorem to evaluate
$\oint_C (x^2y \, dx + y^3 dy)$ where C is the closed path formed by y=x and y= x^3 from (0,0) to (1,1). (7M+7M)

b) Evaluate by Stokes' theorem $\oint_C [(x+y)dx + (2x-z)dy + (y+z)dz]$ where C is the boundary of the triangle with vertices (0,0,0), (1,0,0) and (1,1,0)

[B17 BS 1203]

[B17 BS 1204]

I B. Tech II Semester (R 17) Regular Examinations

ENGINEERING PHYSICS

(Common to CIV,EEE & ME)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT – I

1. (a) Describe the interference phenomena In thin transparent films for reflected light and obtain the conditions for maxima and minima. [7M]
   (b)Discuss the Fraunhoffer diffraction of monochromatic light at a single slit. [7M]

(OR)

2. (a)Describe, with a neat sketch, the design and working of Michelson's interferometer. [7M]
   (b)Explain how the resolving power of a grating can be determined. [7M]

UNIT – II

3. (a)Differentiate spontaneous and stimulated emission processes and obtain the Einstein's relation for spontaneous to stimulated emission coefficients. [7M]
   (b) Define numerical aperture of an optical fiber and derive an expression for the same. [7M]

(OR)

4. (a)With neat sketches, explain the principle and working of He – Ne gas laser system. [7M]
   (b)Explain the characteristics of lasers and mention the applications of lasers. [7M]

UNIT – III

5. (a)Discuss the electric fields induced due to time varying magnetic fields and deduce the Faraday's law. [7M]
   (b)Describe any one method of detecting ultrasonics and mention the important applications of Ultrasonics. [7M]

(OR)

6. (a) Explain the concept of displacement current, and describe the significance of Maxwell's equations. [7M]
   (b)What is magnetostriction and describe the magnetostriction method of producing Ultrasonics. [7M]

UNIT – IV

7. (a)What are matter waves and describe an experiment confirming the wave nature of electrons. [7M]
   (b)What are the salient features of Kronig - Penny model. [7M]

(OR)

8. (a)Explaning the physical significance of wave function of a particle derive the Schrodinger's time independent wave equation. [7M]
   (b)Using band theory of solids how do you classify the materials. [7M]

UNIT - V

9. (a)What is a unit cell and describe the different crystal systems possible in solids. [7M]
   (b)What are nano materials and explain the chemical vapour deposition method of fabricating nano materials. [7M]

(OR)

10. (a) Define packing fraction and deduce the packing fraction for a simple cubic structure. [7M]
    (b)Define the basic approaches of fabricating nano materials and discuss the sol-gel method. [7M]

[B17 BS 1204]

[B17 BS 1205]

I B. Tech II Semester (R 17) Regular Examinations

ENGINEERING CHEMISTRY

(Common to CSE,ECE & IT)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT - I

1. (a) Explain the mechanism of free radical Polymerization reaction with a suitable example. [7M]

   (b)Distinguish between thermoplastic and thermosetting resins. [7M]

(OR)

2. (a)What are conducting Polymers? Discuss the applications of conducting Polymers. [7M]

   (b)Write notes on Bu Na – S and Bu Na – N. [7M]

UNIT - II

3. (a) Explain the Proximate analysis of coal and give its significance. [7M]

   (b) Explain the fractional distillation of crude oil. [7M]

(OR)

4. (a)Write notes on (i) Knocking (ii) CNG [7M]

   (b)How Synthetic Petrol can be prepared by Bergius Process**.** [7M]

UNIT - III

5. (a) Explain the mechanism of electrochemical theory of corrosion with neat diagram. [7M]

   (b) Describe briefly about cathodic Protection. [7M]

(OR)

6. (a) Explain Hydrogen – Oxygen fuel cell with neat cell diagram [7M]

   (b)Discuss on various constituents of Paint. [7M]

UNIT - IV

7. (a) What is hardness? How it is determined by EDTA method? Explain. [7M**]**

   (b) Describe with equations how water can be softened using Lime & Soda Process [7M]

(OR)

8. (a) Discuss various sterilizing methods used in municipal water treatment. [7M]

   (b) Illustrate the reverse osmosis Process with a neat diagram. [7M]

UNIT - V

9. (a) Discuss chemistry involved in setting and hardening of cement? [7M]

   (b) What are refractories? Discuss the classification of refractories . [7M]

(OR)

10. (a) Write the engineering applications of Liquid Crystals . [7M]

    (b) Explain the stoichiometric defects in crystals. [7M]

[B17 BS 1205]

[B17 CS 1201]

I B. Tech II Semester (R 17) Regular Examinations

COMPUTER PROGRAMMING USING C

(Common to CIV,EEE & ME)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.

*****

UNIT-I

1. a) Discuss about computer languages. [7M]
   b) Explain c tokens. [7M]

(OR)

2. a) Explain different c operators. [7M]
   b) write about algorithm, pseudo code and flowchart. [7M]

UNIT-II

3. a) Discuss various looping techniques in c. [7M]
   b) Write a c program for summation of n numbers. [7M]

(OR)

4. a) Explain 2-D arrays and character arrays in c. [7M]
   b) Write a c program to find frequency of characters of a string. [7M]

UNIT-III

5. a) Explain parameter passing techniques in c. [7M]
   b) Write a c program for towers of Hanoi using recursive function. [7M]

(OR)

6. a) Explain storage classes in c. [7M]
   b) Write a c program for Fibonacci series using recursive function. [7M]

UNIT-IV

7. a) What is a pointer? How pointer variables are initialized. [7M]
   b) Write a program to print command line arguments on the screen. [7M]

(OR)

8. a) Discuss character pointers with examples. [7M]
   b) Write a c program to pass pointer variables as function arguments. [7M]

UNIT-V

9. a) Explain the difference between structure and union and write a program to find sum of marks in 3 subjects for a student using structures. [7M]
   b) Explain different bit-wise operators in c. [7M]

(or)

10. a) Explain about the input and output operations of a file. [7M]
    b) Write a c program to open a file and to print its contents on screen. [7M]

[B17 CS 1201]

[B17 CS 1202]

I B. Tech II Semester (R 17) Regular Examinations

OBJECT ORIENTED PROGRAMMING THROUGH C++

(COMPUTER SCIENCE & ENGINEERING)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. A. What are the features of object programming language? [7M]
   B. List the drawbacks of conventional programming? [7M]

(OR)

2. A. Explain array of objects with a suitable program? [7M]
   B. Explain inline function with an example? [7M]

UNIT-II

3. A. Explain constructor overloading with an example? [7M]
   B. Explain assignment overloading with a suitable example? [7M]

(OR)

4. A. Explain Dynamic initialization of Objects? [7M]
   B. What is operator overloading? Write a C++ program illustrating overloading binary operators? [7M]

UNIT-III

5. A. Explain the concepts of pointers to objects? [7M]
   B. What is virtual base class? Write a C++ program illustrating virtual base classes? [7M]

(OR)

6. A. Explain virtual function with an example? [7M]
   B. Explain hybrid inheritance with an example? [7M]

UNIT-IV

7. A. What is an Exception? Explain about try, throw and catch with example? [7M]
   B. Explain unformatted I/O operations with examples? [7M]

(OR)

8. A. Explain the principles of exception handling? [7M
   B. What are the String Characteristics? [7M]

UNIT-V

9. A. Explain about different types of containers? [7M]

   B. Write a program for bubble sort using function templates? [7M]

(OR)

10. A. Explain the concepts of command line arguments. [7M]

    B. Explain differences between templates and macros? [7M]

[B17 CS 1202]

[B17 CS 1203]

I B. Tech II Semester (R 17) Regular Examinations
DATA STRUCTURES
(Electronics and Communication Engineering)
MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
*****

UNIT-I

1. A) Define data structure. Discuss different types of data structure their implementations applications. (7M)
   B) Implement binary search technique using recursion. (7M)

(OR)

2. A) What is an array? Discuss different types of array with examples. (7M)
   B) Rearrange following numbers using quick sort: (7M)
   10, 6, 3, 7, 17, 26, 56, 32, 72

UNIT-II

3. A) Write an algorithm for basic operations of stack. (7M)
   B) Explain the procedure to evaluate postfix expression. Evaluate the following postfix expression 7 3 4 + - 2 4 5 / + * 6 / 7 +? (7M)

(OR)

4. A) Define Queue. Explain the operations of queue using arrays. (7M)
   B) Explain the advantages of circular queue (7M)

UNIT-III

5. A) Define pointer. Explain Dynamically allocated storage using pointers. (7M)
   B) Write an Algorithm for the operations of Linked stack (7M)

(OR)

6. A) Write an Algorithm for the operations of single Linked list (7M)
   B) Explain polynomial addition using Linked List (7M)

UNIT-IV

7. A) What is a Binary tree. Explain threaded binary tree. (7M)
   B) Explain Binary tree traversal techniques. (7M)

(OR)

8. A) Explain the operations of Binary search trees. (7M)
   B) Define Max Heap. Write an algorithm for deletion of elements from Max Heap. (7M)

UNIT-V

9. A) What is a graph? Explain the properties of graphs. (7M)
   B) Write breadth first traversal algorithm. Explain with an example. (7M)

(OR)

10. A) Define Minimum spanning tree. Explain Kruskal's Algorithm. (7M)
    B) Write an Algorithm to find shortest path in a Graph (7M)

[B17 CS 1203]

[B17 ME 1201]

I B. Tech II Semester (R 17) Regular Examinations

ENGINEERING DRAWING

(Common to CSE,ECE & IT)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

Assume the missing data if any, suitably

*****

UNIT-I

1. An inelastic string 145 mm long has its one end attached to the circumference of a circular disc of 40 mm diameter. Draw the curve traced out by the other end of the string, when it is completely wound around the disc, keeping the string always tight. [14 M]

[OR]

2. Two fixed points A and B are 100mm apart, Trace the complete path of a point P moving (in the same plane as that of A and B) in such a way that the sum of its distance from A and B is always the same and equal to 125mm. Name the curve and draw another curve parallel to and 25mm away from this curve. [14 M]

UNIT-II

3. A line AB, of 80 mm long has its end **A**, 15 mm in front of VP and 20 mm above HP. The other end B is 40 mm above HP and 50 mm in front of VP. Draw the projections of the line and determine the inclinations of the line with HP and VP. [14 M]

[OR]

4. (a) Draw the projections of a 75mm long straight line in the following positions: (i) parallel to and 30mm above the HP and in the VP; (ii) perpendicular to the VP, 25mm above the HP and its one end in the VP; (iii) Inclined at 30^0 to the HP and its one end 20mm above it, parallel to and 30mm in front of the VP. [7 M]

(b) Draw the projections of the following points on the same ground line, keeping the projectors 25mm apart. (i) Point A in the HP and lying 20mm behind the VP; (ii) Point B is 40mm above the HP and 25mm in front of the VP; (iii) Point C is 25mm below the HP and 25mm behind the VP; (iv) Point D is 15mm above the HP and 50mm behind the VP. [7 M]

UNIT-III

5. Draw a rhombus of diagonals 100 mm and 60 mm long, with the longer diagonal horizontal. The figure is the top view of a square of 100mm long diagonals, with a corner on the ground. Draw its front view and determine the angle which its surface makes with the ground. [14 M]

[OR]

6. A semicircular plate of 40mm diameter has its straight edge in the VP and inclined at 45^o to the HP, the surface of the plate makes an angle of 30^o with the VP. Draw its projections. [14 M]

UNIT-IV

7. A hexagonal pyramid, base 25mm side and axis 50mm long, has an edge of its base on the ground. Its axis is inclined at 30^0 to the ground and parallel to the VP. Draw its projections. [14 M]

[OR]

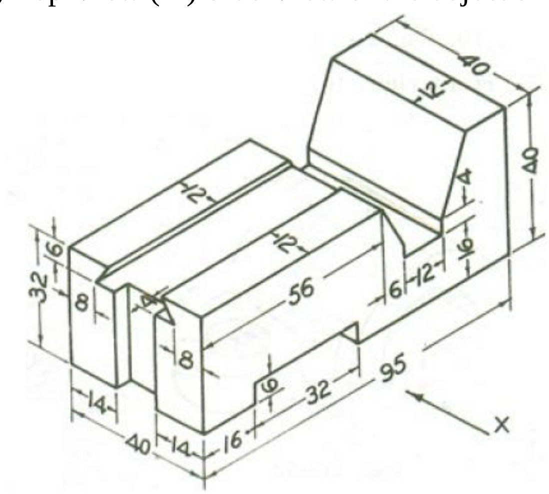
8. Draw the projections of a cylinder 75mm diameter and 100mm long, lying on the ground with its axis inclined at 30^o to the VP and parallel to the ground. [14 M]

UNIT-V

9. A square pyramid with base side 40mm and height 60mm is resting on a cube of sides 50mm, the axes of the cube and the pyramid being in the same line. Two sides of the base of the pyramid are parallel to the edges of the cube. Draw the isometric view. [14 M]

[OR]

10. Draw (i) Front View (ii) Top View (iii) Side View of the object shown below: [14 M]



All the dimensions are in mm

[B17 ME 1201]

[B17 CE 1201]

I B. Tech II Semester (R 17) Regular Examinations

BUILDING MATERIALS AND CONSTRUCTION
(For Civil)
MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
****

UNIT-I

1. a) Explain the classification of bricks. (Including IS classification) [7 M]

   b) Explain the following clay products: [7 M]
      i) Stoneware
      ii) Terra-cotta

(OR)

2. a) Explain the term "Quarrying of stones"? [7 M]
   b) Classify tiles and explain them with neat sketches. [7 M]

UNIT-II

3. a) What is seasoning of timber? Explain the defects due to seasoning of timber. [7 M]

   b) Explain the following wood based products: [7 M]
      i) Block Boards
      ii) Particle Boards

(OR)

4. a) List various classifications of plywood. [7 M]
   b) Draw the cross-section of a tree and explain the various details. [7 M]

UNIT-III

5. a) Define Specific gravity, Bulk density and Porosity of aggregates. [7 M]
   b) Explain the manufacturing process of cement by "Dry" process? [7 M]

(OR)

6. a) Explain the term "Bulking of sand"? [7 M]
   b) State and explain various laboratory tests for testing OPC? [7 M]

UNIT-IV

7. a) What are FAL-G blocks and Concrete blocks [7 M]
   b) What are the characteristics of an ideal paint? [7 M]

(OR)

8. a) Explain various closers in Brick masonry with neat sketches? [7 M]

   b) What is a foundation? Explain different types of foundations? [7 M]

UNIT-V

9. a) What is Roofing? Explain Madras terrace Roof? [7 M]

b) Define Form work and explain the different types of form work. [7 M]

(OR)

10. a) What is Scaffolding? Explain the different types of Scaffoldings? [7 M]

b) List out various staircases and explain any two them with neat sketches. [7 M]

[B17 CE 1201]

[B17 EC 1201]

I B. Tech II Semester (R 17) Regular Examinations

ELEMENTS OF ELECTRONICS ENGINEERING
(Common to CSE & IT)
MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
*****

UNIT I

1. a)Explain in detail about drift and diffusion currents. (7M)
   b) Explain Hall Effect and it's applications in detail. (7M)

OR

2. a) What are the differences between Avalanche breakdown andZener Breakdown. (7M)
   b) Explain the basic operation of semiconductor diode with v-I characteristics. (7M)

UNIT II

3. a)Explain the V-I Characteristic ofzenerDiode, and state its applications. (7M)
   b)Explain Tunneling phenomenon and V-I Characteristics of Tunnel diode. (7M)

OR

4. a) Derive expression for the ripple factor and efficiency of half wave rectifier without filter. (7M)
   b) With aneat diagram, explain the operation of full wave rectifier and obtain expression for with filter Ripple factor. (7M)

UNIT III

5. a) Plot the input and output characteristics of transistor in CE configuration and explain the shape of the characteristics. (7M)
   b) What is early effect and what are its consequences. (7M)

OR

6. a). Plot the input and output characteristics of the transistor in CB configuration and explain shape of the curves. (7M)
   b) Define α and β.Derive the relation between α and β. (7M)

UNIT IV

7. a)Draw and explain the Self biasingcircuit. Derive an expression for Stability factor S. (7M)
   b) Explain the phenomenon of Thermal runaway. (7M)

OR

8. a) Explain any two types of bias compensation. (7M)
   b) What are the reasons for the instability of operating point? Briefly explain the methods of stabilization of operating point. (7M)

UNIT V

9. a) Explain the operation of JFET with Drain and Transfer characteristics. (7M)
   b) What are the differences between JFET andBJT? (7M)

OR

10. a) Explain the operation of Enhancement MOSFET in details. (7M)
    b) Define $\mathbf{g_m}$,$\mathbf{r_d}$and $\boldsymbol{\mu}$ of JFET and give the relation between them. (7M)

[B17 EC 1201]

[B17 EE 1201]

I B. Tech II Semester (R 17) Regular Examinations

CIRCUIT THEORY

(Electrical Electronics Engineering)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. a) Obtain the expressions for star-delta and delta-star equivalence of resistive network. (7M)

   b) Find the value of resistance R, if the current is I=11 A and source voltage is 66 V as shown in figure. (7M)

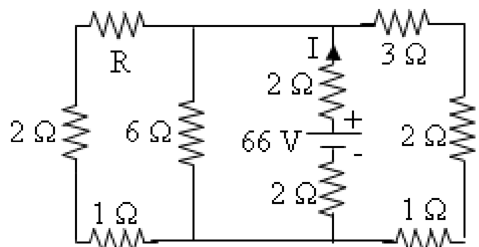


Figure:1

(OR)

2. a) Explain Source Transformation with suitable examples. (7M)

   b) Use the nodal analysis to determine voltage at node 1 and the power supplied by the dependent current source in the network shown in figure.

(7M)

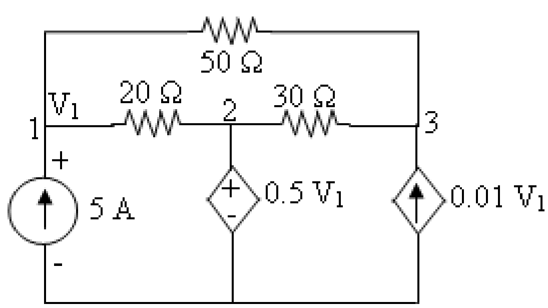


Figure:2

UNIT-II

3.a) Explain the procedure for obtaining fundamental tie-set matrix of given network. (7M)

   b) Draw the oriented graph of a network with fundamental cut-set matrix as shown below.(7M)

Twigs				Links		
1	2	3	4	5	6	7
1	0	0	0	-1	0	0
0	1	0	0	1	0	1
0	0	1	0	0	1	1
0	0	0	1	0	1	0

Also find number of cut-sets and draw them.

(OR)

4.a) For the network graph shown in figure, draw all possible trees. For any one of these trees, prepare a cut-set schedule and obtain the relation between tree-branch voltages and branch voltages. (7M)

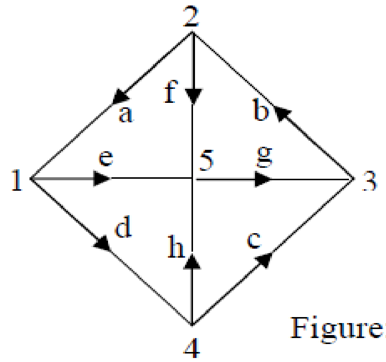


Figure:2

b) Describe the procedure to construct the dual of a network with an example. (7M)

UNIT-III

5.a) A ring has a mean diameter of 21 cm and cross sectional area of 10 cm^2. The ring is made up of semi-circular sections of cast iron and cast steel with each joint having reluctance equal to an air gap of 0.2 mm. Find the ampere turns required to produce a flux of 0.8 milli Wb. The relative permeability of cast steel and cast iron are 800 & 166 respectively. Neglect fringing and leakage effects. (7M)

b) Two identical coupled coils have an equivalent inductance of 80 mH when connected series aiding and 35 mH in series opposing. Find L_1, L_2, M and K. (7M)

(OR)

6. a) Derive the relationship between Flux,MMF and Reluctance. (7M)

b)A coil is wound uniformly with 400 turns over an iron ring having a mean circumference of 50 cm and a cross section of 0.4 cm^2. If the coil has resistance of 10Ω and is connected across a 50V DC supply, calculate the m.m.f of the coil, magnetic field strength, magnetic field density,total flux and reluctance of the ring. (7M)

UNIT-IV

7. a)Define the following: (7M)

i) Amplitude of an alternating quantity

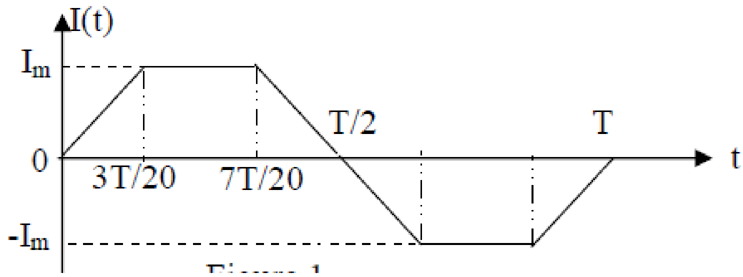
ii) Instantaneous value of an alternating quantity

iii) Frequency

b) Show that power consumed in a purely inductive circuit is zero when sinusoidal voltage is applied across it. (7M)

(OR)

8. a) Find the average value, r.m.s value, form factor and peak factor for the wave form shown in figure. (7M)



b) A coil of inductance L and resistance R in series with a capacitor is supplied at a constant voltage from a variable frequency source. If the frequency is $\dot{\omega}_r$, find in terms of L, R and $\dot{\omega}_r$ the values of those frequencies at which the circuit current would be half as much as that at resonance.Hence or otherwise determine the bandwidth and selectivity of the circuit. (7M)

UNIT-V

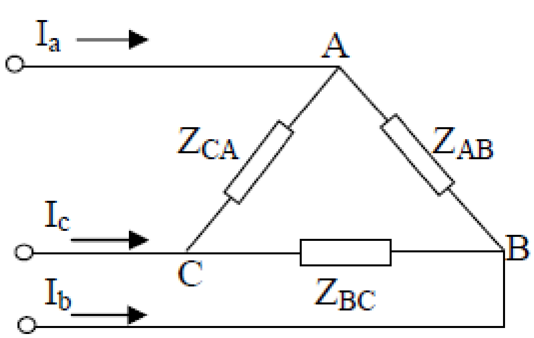
9. a) Explain the relationship between line and Phase quantities in delta connected circuits? (7M)

b) A balanced star connected load of (4+j3) Ω per phase is connected to a balanced 3¢ 400v supply. The phase current is 12 A. Find a) active power b) reactive power c) Apparent power (7M)

(OR)

10. a) A four-wire star-star circuit has Van = $120\angle 120^0$, Vbn = $120\ \angle 0^0$ Vcn = $120\ \angle -120^0$ V. If the impedances are Zan = $20\angle\ 60^0$, Zbn= $30\angle\ 0^0$ and Zcn = $40\ \angle 30^0$ Ω, find the current in the neutra line. (7M)

b) For the circuit shown in figure 3, the line voltage is 240 V. Take Vab as reference and determine following: i) phase currents, ii) line currents, iii) total power absorbed in the load. Also draw Phasor diagram (7M)



$Z_{AB} = 25\Omega$

$Z_{BC} = 12\angle 60^0\,\Omega$

$Z_{CA} = 16\angle -30^0\,\Omega$

[B17 EE 1201]

[B17 EE 1202]

I B. Tech II Semester (R 17) Regular Examinations

BASIC ELECTRICAL & ELECTRONICS ENGINEERING

(Mechanical Engineering)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

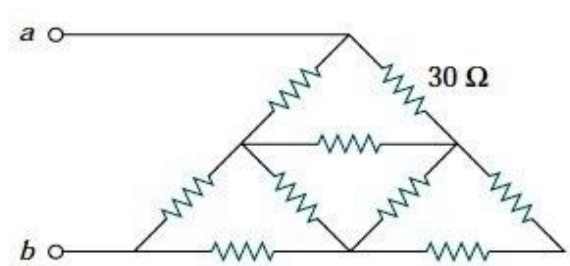
All questions carry equal marks.

*****

UNIT-I

1. (a) State and explain Kirchhoff's Laws with example. [7M]

   (b) Find the equivalent resistance Rab for the circuit shown below. All the resistor Values are 30Ω. [7M]



OR

2. (a) Define Dynamically Induced E.M.F and derive expression for it. [7M]

   (b) A coil having an inductance 60mH is carrying a current of 60A. Calculate the Self-induced EMF in the coil. When the current in the coil reversed in 30milliseconds. [7M]

UNIT-II

3. (a) Derive the EMF equation of DC generator [7M]

   (b) A shunt generator supplies a load of 7.5KW at 200V, Calculate the generated emf if armature resistance is 0.6Ω and field resistance of 80Ω. [7M]

OR

4. (a) Derive the torque equation of the DC motor. [7M]

   (b) An 8-pole, wave-connected armature has 600 conductors and is driven at 625 rev/min. If the flux per pole is 20 mWb, determine the generated E.M.F. [7M]

UNIT-III

5. (a) Derive the EMF equation of a single phase transformer. [7M]

(b) A 200 KVA rated transformer has a full-load copper loss of 1.5 kW and an iron loss of 1 kW. Determine the transformer efficiency at full load & half load for 0.85 power factor. [7M]

OR

6. (a) Explain the operation of Transformer under NO-LOAD with phasor diagram. [7M]

(b) An ideal 25KVA Transformer has 500 turns on primary and 40 turns on the secondary winding. The primary winding is connected to 3000 V, 50Hz supply. Calculate (i) Primary and secondary currents (ii) Secondary EMF (iii) Maximum flux. [7M]

UNIT-IV

7. (a) Draw and explain the slip-Torque Characteristics of Three phase Induction motor. [7M]

(b) The frequency of the supply to the stator of a 6-pole induction motor is 50 Hz and the rotor frequency is 2 Hz. Determine (i) the slip, and (ii) the rotor speed in r.p.m [7M]

OR

8. (a) Derive the EMF equation of Alternator [7M]

(b) Obtain the Voltage Regulation of Alternator by synchronous impedance method [7M]

UNIT-V

9. a) Explain the operation of Diode in Forward and reverse bias conditions and draw V-I characteristics. [7M]

(b) Explain the operation of Zener diode and draw its V-I characteristics [7M]

OR

10. (a) Draw the circuit diagram of Bridge rectifier and explain its operation. [7M]

(b) Explain how the transistor acts as an amplifier. [7M]

[B17 EE 1202]

[B17 EE 1203]

I B. Tech II Semester (R 17) Regular Examinations

ELEMENTS OF ELECTRICAL ENGINEERING

(Electronics and Communication Engineering)

MODEL QUESTION PAPER

Time: 3 hours **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.

*****

UNIT-I

1. (a) State and explain Kirchhoff's Laws with example. [7M]

   (b) Derive star-delta and delta- star Transformation for Equal resistances [7M]

(OR)

2. (a) Define Dynamically Induced E.M.F and derive expression for it. [7 M]

   (b) A coil having an inductance 60mH is carrying a current of 60A. Calculate the Self-induced EMF in the coil. When the current in the coil reversed in 30milliseconds. [7M]

UNIT-II

3. (a) Derive the EMF equation of DC generator [7M]

   (b) A series motor drives a load at 1500 r.p.m and takes a current of 20A when the supply voltage is250V if the total resistance of the motor is 1.5 ohms and the iron, friction and windage losses amount to 400W. Determine the efficiency of the motor. [7M]

(OR)

4. (a) Derive the Torque equation of DC motor. [7M]

   (b) A shunt generator supplies a load of 7.5KW at 200V, Calculate the generated emf if armature resistance is 0.6Ω and field resistance of 80Ω. [7M]

UNIT-III

5. (a) Explain the operation of Transformer under NO-LOAD with phasor diagram. [7M]

   (b) An ideal 25KVA Transformer has 500 turns on primary and 40 turns on the secondary winding. The primary winding is connected to 3000 V, 50Hz supply. Calculate (i) Primary and secondary currents (ii) Secondary EMF (iii) Maximum flux. [7M]

(OR)

6. (a) Derive the EMF equation of a single phase transformer. [7M]

(b) A 25-kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000-V, 50-Hz supply. Find the full-load primary and secondary currents, the secondary e.m.f. and the maximum flux in the core. Neglect leakage drops and no-load primary current [7M]

UNIT-IV

7 (a) Explain the Slip - Torque Characteristics of Three phase Induction Motor. [7M]

(b) A 3-Phase Induction Motor is Running at 5% slip. The Output is 36.75KW and Total Mechanical losses are 1.5KW. Estimate the copper losses in the rotor. If the stator losses are 4KW, estimate the efficiency of the Motor. [7M]

(OR)

8**.** (a) Define Slip and Rotor Frequency in Detail. [7M]

(b) The Power Input to 3-φ Induction motor is 55Kw.Total stator losses Equal to 2.2Kw.Find (i) Rotor copper loss (ii) Mechanical Power developed if the motor is running at a speed of 720rpm at 50Hz supply with 4poles. [7M]

UNIT-V

9**.** (a) Derive the EMF equation of Alternator [7M]

(b) Obtain the Voltage Regulation of Alternator by SYNCHRONOUS IMPEDENCE METHOD. [7M]

(OR)

10**.** (a) Explain the operation of PMMC with neat sketches [7M]

(b) Explain Deflecting, controlling and damping Torques with neat sketches [7M]

[B17 EE 1203]

[B17 BS 2101]

II B. Tech I Semester (R 17) Regular Examinations

MATHEMATICS-IV

Common to CE,ECE,EEE & ME

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a). Prove that the function defined by 7M

$f(z) = \dfrac{(1+i)x^3 - (1-i)y^3}{x^2+y^2} (z \neq 0)$ *and* $f(0) = 0$ is continuous and the Cauchy-Riemann equations are satisfied at the origin, yet $f'(0)$ do not exist.

(b). Find the complex potential w and velocity potential ϕ when its stream function is $\psi = \tan^{-1}\left(\dfrac{y}{x}\right)$ 7M

(OR)

2. (a). If $f(z)$ is a regular function of z, prove that 7M

$\left(\dfrac{\partial^2}{\partial x^2} + \dfrac{\partial^2}{\partial y^2}\right)| f(z)|^2 = 4| f'(z)|^2$

(b). Find the bi-linear transformation which maps the points $Z = 1, -1, \infty$ onto the points $w = 1+i, 1-i, 1.$ Hence find (a) the critical points and (b) the invariant points of this transformation. 7M

UNIT-II

3. (a). Solve $\dfrac{\partial u}{\partial x} = 4\dfrac{\partial u}{\partial y}$, given that $u = 3e^{-y} - e^{-5y}$ when $x = 0$ 7M

(b). A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is vibrating by giving to each of its points a velocity $\lambda x(l-x)$, find the displacement of the string at any distance x from one end at any time t. 7M

(OR)

4. (a). A homogeneous rod of conducting material of length 100 cm has its ends kept at zero temperature and the temperature initially is 7M

$u(\text{x},0) = x, \quad 0 \le x \le 50$

$= 100 - x, \quad 50 \le x \le 100$

Find the temperature $u(x,t)$ at any time.

(b). Solve $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ for $0 < x < \pi,\ 0 < y < \pi$ with conditions given $u(x,\pi) = 0,$ $u(0,y) = 0,\ u(\pi,y) = 0,\ u(\text{x},0) = sin^2 x.$ 7M

UNIT-III

5. (a). Form the difference equation generated by $y_n = (A+Bn)3^n$ 7M

(b). Solve the difference equation $y_{n+2} + y_{n+1} - 56y_n = 2^n(n^2-3)$ 7M

(OR)

6. (a). If $Z(u_n) = \frac{z}{z-1} + \frac{z}{z^2+1}$ find the Z-transform of u_{n+2} 7M

(b). Using Z-transforms, solve $u_{n+2} - 2u_{n+1} + u_n = 3n+5$. 7M

UNIT-IV

7. (a). Twenty identical coins each with probability P of showing heads are tossed. The Probability of heads showing on 10 coins is same as that of heads showing on 11 coins. Find P. 7M

(b). X is a normal variate with mean 30 and standard deviation 5. Find the probability that (i) $26 \le X \le 40$ (ii) $X \ge 45$ (iii) $|X-30| > 5$ 7M

(OR)

8. (a). Derive moment generating function of Poisson distribution. 7M

(b). In a Normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. 7M

UNIT-V

9. (a). A sample of height of 6400 soldiers has a mean of 67.85 inches and a standard deviation of 2.56 inches while a simple sample of heights of 1600 sailors has a mean of 68.55 inches and a standard deviation of 2.52 inches. Do the data indicate that the sailors are on the average taller than soldiers? 7M

(b). A die is tossed 960 times and it falls with 5 upwards 184 times. Is the die biased? 7M

(OR)

10. (a). A sample of 10 measurements of the diameter of a sphere gave a mean of 12 cm and a Standard deviation of 0.15 cm. Find 95% confidence limits for the actual diameter. 7M

(b). The number of accidents in a street during a particular week is given below: 7M

Days:	Mon	Tue	Wed	Thu	Fri	Sat
No. of accidents:	5	4	3	7	6	5

Test the hypothesis that the number of accidents does not depend on the day of the week.

[B17 BS 2101]

[B17 CE 2101]
II B. Tech I Semester (R 17) Regular Examinations
MECHANICS OF SOLIDS
CIVIL ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
*****

UNIT-I

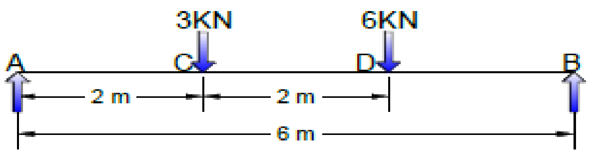
1. (a). Explain the Stress-Strain diagram for mild steel. 7M
   (b). A steel rod of 3cm diameter and 5m long is connected to two grips and the rod is maintained at a temperature of 95˚C. Determine the stress exerted when the temperature falls to 30˚C if i) the ends do not yield
   ii) the ends yield by 0.12 cm 7M
   Take E= $2x10^5$ MN/m^2 α= $12x10^{-6}$/˚C

(OR)

2. (a). Derive the strain energy stored in a body when the load is gradually applied. 7M
   (b). A steel rod is 2m long and 50 mm in diameter. An axial pull of 100KN is suddenly applied to the rod. Calculate the instantaneous stress induced and also the instantaneous elongation produced in the rod. Take E=200 GN/m^2 7M

UNIT-II

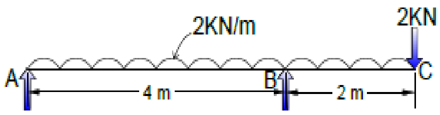
3. (a). A Simply supported beam of length 6m, carries a point load of 3 KN and 6KN at distances of 2m and 4m from the left end. Draw SFD and BMD. 7M



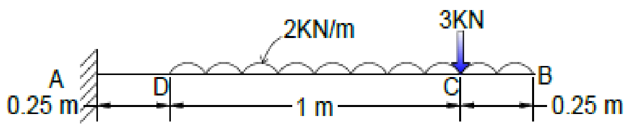
   (b). Derive the relation between Load , Shear Force and Bending Moment. 7M

(OR)

4. (a). Draw SFD and BMD for the Over Hanging beam carrying UDL of 2 KN/m over the entire length and a point load of 2KN as shown in figure. Locate the Point Of Contraflexure. 7M



   (b). A Cantilever 1.5m long is loaded with a UDL of 2KN/m run over a length of 1.25m from free end. It also carries a point load of 3KN at a distance of 0.25m from free end. Draw SFD and BMD . 7M



UNIT-III

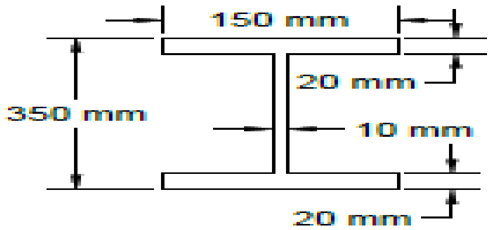
5. (a). A steel plate of width 120 mm and of thickness 20mm is bent in to a circular arc of radius 10m. Determine the maximum stress induced and the bending moment which will produce the maximum stress. Take $E=2X10^5$ N/mm^2. 7M

(b). A rectangular beam 200mm deep and 300mm wide is Simply Supported over a span of 8m. What uniformly distributed load per metre the beam carry, if the bending Stress is not to exceed 120N/mm^2. 7M

(OR)

6. (a). Define Section Modulus. Find an expression for Section Modulus for a Circular and Hollow Circular Sections. 7M

(b). An I-Section beam 350mmx150mm has a web thickness of 10mm and a flange thickness of 20mm. If the Shear force acting on the section is 40KN. Find the maximum shear stress developed in the I-Section. 7M



UNIT-IV

7. (a). Derive an expression for the Normal and Tangential Stresses on an oblique plane, when the body is subjected to direct stresses in two mutually perpendicular direction accompanied by a shear stress. 7M

(b). At a point within a body is subjected to two mutually perpendicular directions, the stresses are 150 N/mm^2(tensile) and 100 N/mm^2(compression). Each of the above stresses is accompanied by a shear stress of 100N/mm^2. Determine the Normal, Shear and Resultant stresses on an oblique plane inclined at 45° with the axis of minor tensile stress 7M

(OR)

8. (a). A solid shaft has to transmit 75KW at 200 rpm, taking allowable shear stress as 70N/mm^2. Find suitable diameter for shaft, if the maximum torque transmitted at each revolution exceeds the mean by 30%. 7M

(b). A Closely Coiled helical spring of round steel wire 10mm in diameter having 10 complete turns with a mean diameter of 12cm is subjected to an axial load of 200N. Determine 7M

i) Deflection of Spring.

ii) Maximum Shear Stress in the wire.

iii) Stiffness of Spring. Take $C= 8x10^4$ N/mm^2

UNIT-V

9.	(a).	Write about Prof. Perrys Formula.	7M
	(b).	A 1.5m long column has a circular cross section of 5cm diameter. One of the ends of the column is fixed in direction and position and other end is free. Taking Factor Of Safety as 3. Calculate the safe load using: i) Rankines formula, take yield stress =560N/mm^2 & α=1/1600 ii) Eulers formula ,Youngs Modulus for cast Iron= 1.2x10^5 N/mm^2	7M
		(OR)	
10.	(a).	Derive an expression for Crippling load when both the ends of the column are hinged.	7M
	(b).	A hollow cast iron column 200mm outside diameter and 150mm inside diameter, 8m long has both ends fixed. It is subjected to an axial compressive load. Taking a Factor Of Safety=6, Yield Stress=560 N/mm^2,& α=1/1600 Determine Safe Rankines load.	7M

[**B17 CE 2101**]

[B17 CE 2102]

II B. Tech I Semester (R 17) Regular Examinations

ENVIRONMENTAL ENGINEERING – I

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a). What is meant by "per capita demand" of water? Explain how do you estimate water demand for a town or city — 7M
   (b). Explain the role of an Environmental Engineer in society — 7M

(OR)

2. (a). Using any two methods, calculate the population of a city given below for 2021 and 2031. — 7M

Year	1981	1991	2001	2011
Population	1,10,000	1,55,000	2,06,000	2,42,000

   (b). Explain the importance of protected water supply system for public — 7M

UNIT-II

3. (a). What is meant by Infiltration gallery? Explain with a neat sketch, its construction and operation. — 7M
   (b). Distinguish between surface water, open well and deep bore wells water with respect to suitability, quantity and quality of water. — 7M

(OR)

4. (a). How is the capacity of storage reservoir determined? — 7M
   (b). Derive expressions for determining yield from wells penetrating unconfirmed and confirmed aquifers. — 7M

UNIT-III

5. (a). What is an intake structure? What are the factors governing the selection of intake structure? — 7M
   (b). Explain laying of pipelines, jointing and testing. — 7M

(OR)

6. (a). Explain the harmful effects of excess turbidity, chlorides and nitrates in water. — 7M
   (b). Mention different types of pipes and pipe materials and write the merits and demerits. — 7M

UNIT-IV

7. (a). Explain construction, operation and maintenance of rapid sand filter with neat sketch. — 7M
   (b). Calculate the size of a rectangular sedimentation tank to treat 2 MLD of water. The detention period may be assumed as 4 hours and the overflow rate to be 40,000 litres/m^2 of surface area per day. — 7M

(OR)

8. (a). What are the various chlorination practices and write the significance of break point chlorination. — 7M
   (b). Design a slow sand filter for the given data: — 7M

   Population = 40000
   Rate of water supply = 120 lpcd
   Rate of filtration = 300 l/m^2/day
   L/B ratio = 2:1.

UNIT-V

9.	(a).	What are the different types of layouts of distribution? Describe them in detail with their merits and demerits.	7M
	(b).	Explain the purpose of pumping in a distribution system.	7M
		(OR)	
10.	(a).	Explain the causes and effects of hardness in water. Explain in detail Zeolite method.	7M
	(b).	Explain about the "Nalgonda Technique" for removal of fluorides	7M

[**B17 CE 2102**]

[B17 CE 2103]

II B. Tech I Semester (R 17) Regular Examinations

BUILDING PLANNING AND DESIGN

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer ONE Question from each Unit

Part-A consists of 4 questions, 2 questions for each of Unit –I & II and Part-B consists of a compulsory question for 38 Marks

All parts of the question must be answered in one place only

*****

UNIT-I

1. (a). What is meant by orientation of building? Mention some suggestions for good orientation of the building in a tropical climate. 8M

(b). Differentiate between 8M

i. Privacy and speculation
ii. Aspect and orientation
iii. External privacy and Internal privacy
iv. Horizontal circulation and vertical circulation

(OR)

2. (a). What are the principles of panning of building? Explain with suitable examples the significance of Aspect and Prospect in the planning of a residential building 8M

(b). Define ventilation. What are the systems of ventilation and write the effects of humidity on comfort conditions. 8M

UNIT-II

3. (a). What are the functional requirements of a residential buildings, explain in detail. 8M

(b). What are the different types of residential buildings and briefly explain the selection of site for residential building 8M

(OR)

4. (a). Discuss briefly the guidelines for planning and drawing of residential buildings 8M

(b). Discuss the requirements of a residential building to accommodate a family of 6 members 8M

UNIT-III

5. (a). Draw neat sketches to indicate the conventional signs for the following. 8M

i. Brick masonry
ii. Timber
iii. Concrete
iv. Wash basin

(b). The line plan of a residential building is shown in Figure 1. Draw 30M

i. Dimensioned Plan
ii. Section along AB
iii. Front elevation by following the specifications given below.

Specifications:

1. Foundation: The depth of foundation shall be 1000 mm below ground level. The plain cement concrete (1:4:8) bed in the foundation will be 800 mm wide and 200 mm deep. The footings shall be of brick masonry in C.M (1:4). Width of first and second footing will be 500 mm and 400 mm respectively, where as depth of both footings will be 400M.
2. Basement: The height of basement is 600mm. damp proof course of 50 mm thick shall be provided under the superstructure walls. Thickness of walls in the basement is 300mm.
3. Superstructure: the walls in the superstructure will be of brick masonry in C.M (1:6) and all walls except the partition between toilets are 200 mm thick and the partition walls are 100 mm thick from floor. A square brick pillar 200X200 mm is provided at the left corner in front verandah.
4. Lintels and sunshades: lintels with R.C.C (1:2:4) are provided on all openings and depth is 150mm with a bearing of 150 mm on either side.
5. Sun shades100 mm thick at the wall face and 75 mm thick at free end are provided. All sun shades shall project 600 mm from face of the wall.
6. Height of super structure: The walls in the superstructure are taken to a height of 3300 mm.
7. Roofing: roofing consists of R.C.C (1:2:4) slab 110 mm thick and weather proof course with two courses of flat tiles in C.M (1:4) 50 mm thick is laid over it.
8. Flooring: flooring shall be of polished Shahabad stone slab 25 mm thick over 80 mm thick cement concrete (1:3:6) over sand filling basement.
9. Parapet wall: Parapet wall 100 mm thick and 700 height with brick masonry in C.M (1:4) shall be constructed all around the building.
10. Steps: steps provided in the front and rear side of length 1200 mm. the width of tread is 300mm and rise if step is 150mm.

D_1 1000X2100Panelled door
D_2 900X2000 Panelled door
W_1 1200X1500Glazed window
W_2 1000X1500 Glazed window
V_1 1000X600 Glazed ventilator

Cupboard 1200X1500Flushed shutter

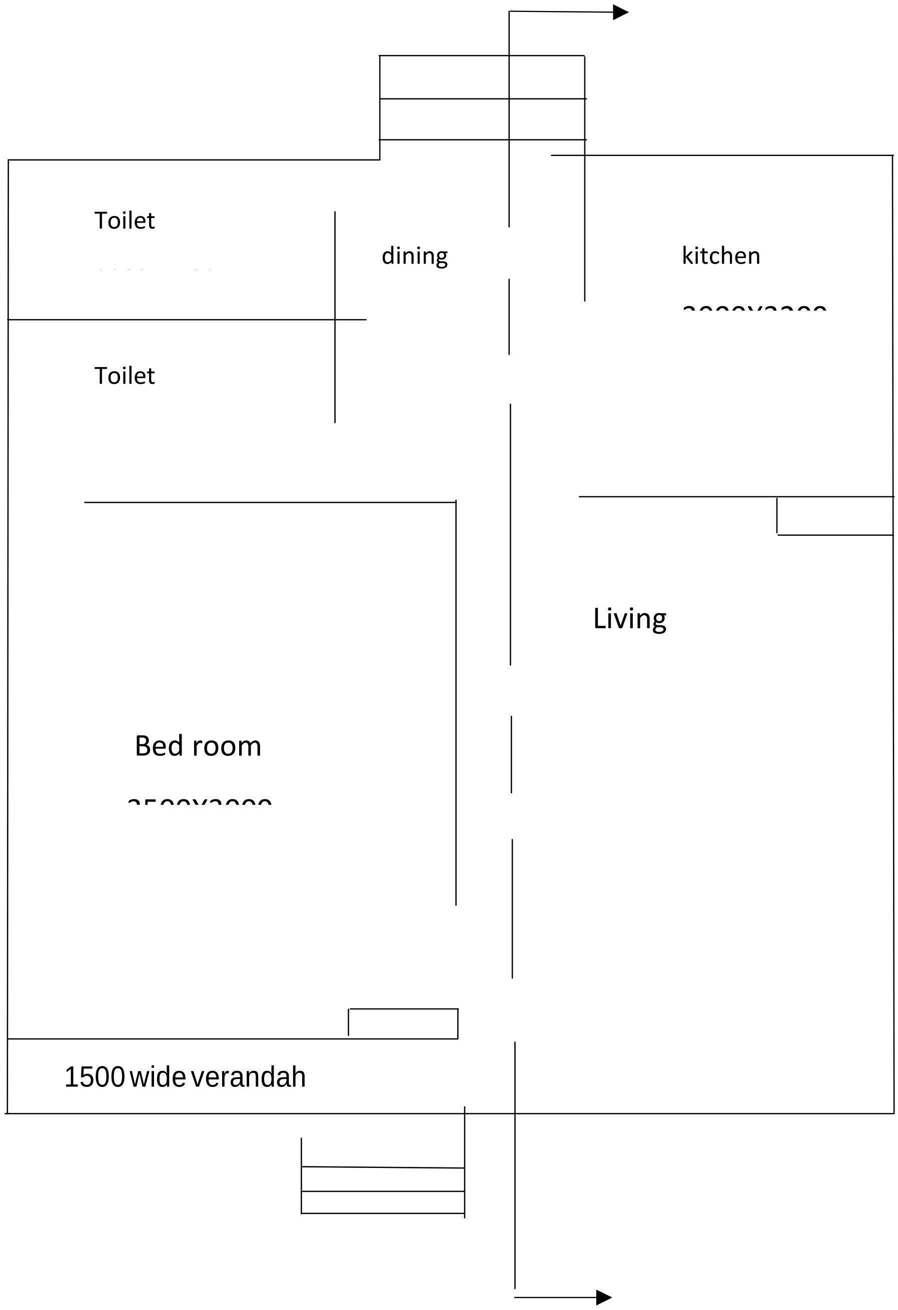


Figure1.

[B17 CE 2103]

[B17 CE 2104]

II B. Tech I Semester (R 17) Regular Examinations

SURVEYING -I

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1.	(a).	What are the primary divisions of survey? Explain.	2M
	(b).	What is a shrunk scale and shrinkage ratio?	2M
	(c).	A steel tape of 20m long standardized at 60°F with a pull of 10kgs was used for measuring a base line. Find the correction for tape length if the temperature at the Time of measurement was 90°F and the pull exerted is 15kgs.weight of 1cubic centimetre steel is 7.86 gram. Weight of tape is 0.8kg.E=2.109x10^6 kg cm^2.coefficient of expansion of tape for1°F=6.2x10^6.Calculate correction for temperature, pull and sag.	10M

(OR)

2.	(a).	What are the principles of surveying?	2M
	(b).	What is the difference between a plan and a map?	2M
	(c).	A 25m chain was found to be 12cms too long after a chaining a distance of 1200m.It was found to be 20cms too long at the end of days work after chaining a total distance of 2500m.find the true distance if the chain was correct before the Commencement of work.	10M

UNIT-II

3.	(a).	Distinguish between true meridian and magnetic meridian	2M
	(b).	Convert the following WCB to QB:22°30’ & 170°12’ Convert the following QB to WCB:N12°24’E & S31°36’E	2M
	(c).	Determine the values of included angles in the closed compass traverse ABCD in the clockwise direction given the following fore bearings of their respective lines.	10M

Line	F.B
AB	40°
BC	70°
CD	210°
DE	280°

Apply the check

(OR)

4.	(a).	What is local attraction? Explain	2M
	(b).	Explain magnetic dip	2M
	(c).	The following bearings were observed in running a complete traverse	10M

LINE	FB	BB
AB	75°5’	254°20’
BC	115°20’	296°35’
CD	165°35’	345°35’
DE	224°50’	44°5’
EF	304°50’	125°5’

At what stations do you suspect the local attraction? Determine the corrected magnetic bearing of declination was 5°10’E.What are the true bearings.

UNIT-III

5. (a). What are the checks in a closed traverse? 2M
   (b). What are the latitude and departure? Give pictorial representation 2M
   (c). For balancing a traverse, give Bowditch graphical method and also axis method. Draw diagrams clearly. 10M

(OR)

6. The following staff readings were observed successively with a level, the instrument having been moved after third, sixth and eighth reading readings 2.228, 1.606, 0.988, 2.090, 2.864, 1.262, 0.602, 1.982, 1.044, 2.644m. Enter the above readings in a page of a level book and calculate R.L of points, if the first was taken with a staff held on a B.M of 432.384m. 14M

UNIT-IV

7. (a). It was required to ascertain the elevations of two points P and Q and a line of level was run from P to Q.The levelling was then continued to a BM of 83.500.The readings obtained are shown below. Obtain the R.L of P and Q. 14M

Station	B.S	I.S	F.S	H.I	R.L	Remarks
1	1.622				P	
2	1.874		0.354			T.P1
3	2.032		1.780			T.P2
4		2.362			Q	
5	0.984		1.122			T.P3
6	1.906		2.824			T.P4
7			2.036		83.50	B.M

Make an arithmetical check.

(OR)

8. (a). What is reciprocal levelling? Explain with diagrams 5M
   (b). How do you calculate the effects of curvature and refraction? 4M
   (c). What is profile levelling? Explain longitudinal sectioning and cross sectioning 5M

UNIT-V

9. (a). What are the contour and contour interval? 3M
   (b). What are the characteristics of contours? 7M
   (c). What are the uses of contour? 4M

(OR)

10. (a). Write short notes on ANY FOUR of the following 14M
    i. Plane table and its accessories.
    ii. Ceylon ghat tracer.
    iii. Pantagraph.
    iv. Planimeter.
    v. Errors in chain survey
    vi. Loose needle method and fast needle method

[B17 CE 2104]

[B17 CE 2105]

II B. Tech I Semester (R 17) Regular Examinations

ENGINEERING GEOLOGY

DEPARTMENT OF CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

		UNIT-I	
1.	**(a).**	Define Engineering Geology. Describe about branches of geology and its importance in Civil Engineering works.	**7M**
	(b).	Define Weathering and explain the types of Weathering. Mention the civil engineering importance of weathering.	**7M**
		(OR)	
2.	**(a).**	Describe the landforms produced by running water and describe about its civil engineering importance.	**7M**
	(b).	Explain hydrological cycle and derive the equation for Darcy's law in ground water movement.	**7M**
		UNIT-II	
3.	**(a).**	What is mineral? How do you identify minerals with help of physical properties of minerals.	**7M**
	(b).	Describe Engineering properties of rocks.	**7M**
		(OR)	
4.	**(a).**	What are igneous rocks? Write about textures, structures and forms of Igneous rocks and write its engineering importance	**7M**
	(b).	Write about feldspars and quartz.	**7M**
		UNIT-III	
5.	**(a).**	Explain Geological time scale.	**7M**
	(b).	Define fault and explain faults and joints and their significance in civil engineering.	**7M**
		(OR)	
6.	**(a).**	Define the term fold? Describe about classification of folds with neat sketches	**7M**
	(b).	Write briefly about major geological formations in India	**7M**
		UNIT-IV	
7.	**(a).**	What is remote sensing? Describe Electromagnetic spectrum .	**7M**
	(b).	Write about seismic method for investing site for selection.	**7M**
		(OR)	
8.	**(a).**	Explain principles of geophysical methods. How do you identify soil profile using electrical Resistivity method?	**7M**
	(b).	Enumerate RS and GIS applications in Civil Engineering	**7M**
		UNIT-V	
9.	**(a).**	Write an essay on the role of engineering geologist in planning, design and construction of civil engineering works	**7M**
	(b).	Geological investigation for tunneling practice.	**7M**
		(OR)	
10.	**(a).**	Describe the geological investigations for dams and reservoirs with neat sketches	**7M**
	(b).	Geological investigations for highways, air fields and railway lines	**7M**

[B17 CE 2105]

[B17 CE 2201]
II B. Tech II Semester (R 17) Regular Examinations
STRUCTURAL ANALYSIS-I
CIVIL ENGINEERING
MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**
All questions carry equal marks.
*****

UNIT-I

1. (a). Explain the moment area method to find the slope and deflection of a loaded beam. 7M
   (b). A pin jointed truss loaded with a single load W = 100kN. If the area of cross section of all members shown in fig.(1) is 1000m^2, what is the vertical deflection of point C? Take E = 200kN mm^2 for all the members. 7M

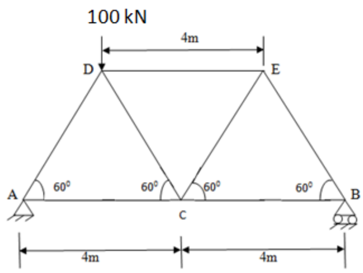


Fig. (1)

(OR)

2. (a). A tension bar 5mis made up of two parts 3m of its length has cross sectional area of 10cm2 while the remaining 2m has a cross sectional area of 20 cm2. An axial load of 80 kN is gradually applied. Find the total strain energy produced in bar and compare this value with that obtained in a uniform bar of same length and having the same volume, when under the same load. Take E = 2 x 105 N/mm2. 7M
   (b). A beam of constant cross section 4m long is free supported at its ends. It is loaded at points 1m from each end with a load of 2kN. Find the ratio of the deflections in the centre of the beam to the deflection at the point under one of the loads. Use conjugate beam method. 7M

UNIT-II

3. (a). Determine the reaction components of a propped cantilever beam s shown in fig.(2). EI is constant throughout. 7M

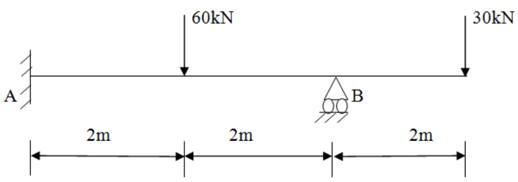


Fig. (2)

   (b). A fixed beam AB of length 6m is subjected to a constructed couple of 500 Nm(clockwise) at a point C which is at a distance of 3.5m from the end A. Draw SF & BM diagram assume "EI" to be constant. 7M

(OR)

4. Analyse the continuous beam shown in fig.(c) using theorem of three moments fig.(3) 7M 14M

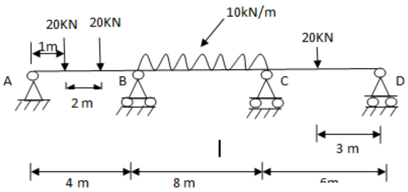


Fig. (3)

UNIT-III

5. A beam ABC is supported at A,B and C and has an internal hinge at D at a distance of 4m from AB. AB = 8m and B = 12m.Draw the influence lines for i) Reaction $R_{A,}$ ii) Reaction R_B and iii) Reaction R_C, iv) Shear at a point just near to B in span BC and v) Moment M_x where X is a point 1m from B in span BC. 14M

(OR)

6. Two point loads of 120 kN and 90 kN, spaced at 2m apart, cross a girder of 15m span, from the left to the right, the smaller load leading. Using influence lines, find the S.F and B.M at mid-span when the leading wheel is 6m from the right hand support. 14M

UNIT-IV

7. The load system shown in fig. (4) crosses a simply supported beam over a span of 24m. Determine (a) the maximum S.F. at a section 8 metres from the left hand end; (b) maximum B.M. under 25,000 N load. 14M

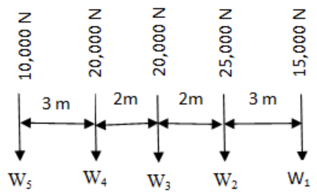


Fig.(4)

(OR)

8. A simple beam of 15m span supporting a uniform dead load 40 kN/m, uniform live load of 60 kN/m, and the moving concentrated loads of 40 kN,100kN,100kN,100kN,100kN and 80kN spaced at 2m,3m,3m,3m and 2m respectively across a girder of 20m span from right to left. Find the max. shear at the left support and at 5m from the left support. Also find out the max. possible moment at the center line of the beam. 14M

UNIT-V

9. (a). Explain different theories of failures. 7M

(b). Find the thickness of the metal necessary for a cylindrical shell of internal diameter 160mm to with stand an internal pressure of 8 N/mm^2. The maximum hoop stress in the section is not to exceed 35 N/mm^2. 7M

(OR)

10. (a). What do you mean by Lame's equations? How will you derive these equations? 7M

(b). A compound cylinder is made by shrinking a cylinder of external diameter 200 mm and internal diameter 160 mm over another cylinder of external diameter 160 mm and internal diameter 120 mm. The radial pressure at the junction after shrinking is 8 N/m^2. Find the final stresses set up across the section, when the compound cylinder is subjected to an internal fluid pressure of 60 N/mm^2. 7M

[B17 CE 2201]

[B17 CE 2202]

II B. Tech II Semester (R 17) Regular Examinations

FLUID MECHANICS- I

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a). What are Dynamic Viscosity and Kinematic Viscosity? 3M
   (b). Explain surface tension and capillarity. 3M
   (c). Calculate the dynamic viscosity of oil which is used for lubrication between a square plate of size 0.8m x 0.8m and an inclined plate with an angle of inclination 30^{O}. The weight of the square plate is 30 Kg (f) and it slides down the inclined plate with a uniform velocity of 0.3 m/s. Thickness of the oil film is 1.5mm. 8M

(OR)

2. (a). Draw a neat sketch of a single column manometer and show how to calculate the pressure head with a single reading. 5M
   (b). A vertical gate closes a horizontal tunnel 5m high and 3m wide running full with water. The pressure at the bottom of the gate is 2x104 kg/m2. Determine the total pressure on the gate and the position of centre of pressure. 9M

UNIT-II

3. (a). What are steady, Uniform, Irrotational and Turbulent flows? 5M
   (b). Derive the continuity equation in Cartesian Coordinates. 9M

(OR)

4. (a). What are stream lines and stream tubes? 2M
   (b). What are basic principles of fluid flow? And what we derive from each principle? 5M
   (c). Two velocity components are given in the following cases, find the third component such that they should satisfy the continuity equation; 7M
   u = x3 + y2 + 2z2 , v = -x2y –yz –xy
   u = log (y2 + z2) , v = log (x2 + z2)

UNIT-III

5. (a). Derive the energy equation by integrating the Euler's Equations of motion. 7M
   (b). A venturimeter has its axis vertical, the inlet and throat diameters being 15cm and 7.5cm respectively. The throat is 22.5cm above inlet and Cd = 0.96. Petrol of specific gravity 0.78 flows up through the meter at a rate of 0.029m3/s. Find the pressure difference in Kg/cm2 between inlet and throat. 7M

(OR)

6. (a). Derive the expression for time of emptying tank with no inflow, through an orifice; 7M

$$t= \frac{2\ A.\sqrt{H_1}}{C_d\ a.\sqrt{2g}}$$

(b). A pipe 300m long has a slope of 1 in 100, tapers from 1.2m diameter at the high end to 0.6m diameter at the low end. Quantity of water flowing is 5400 litres per minute. If the pressure at the high end is 0.70 Kg/cm^2, find the pressure at the low end neglecting the losses. 7M

UNIT-IV

7. A 45^{O} reducing bend is connected in a pipe line, the diameters at the inlet and outlet of the bend being 600mm and 300mm respectively. Find the force exerted by water on the bend, if the intensity of pressure at the inlet to the bend is 8.829 N/cm2 and the rate of flow of water is 600 litres per second. 14M

(OR)

8. (a). In a 45^{O} bend rectangular air duct of 1m2 cross-sectional area is gradually reduced to 0.5m2 area. Find the magnitude and direction of the force required to hold the duct in position, if the velocity of flow at the 1m2 section is 10m/s and pressure is 2.943 N/cm2. Take density of sir as 1.16 Kg/m3 14M

(b). 7M

UNIT-V

9. (a). Derive Darcy - Weisbach equation for head loss due to friction in pipes. 7M

(b). The population of a city is 8,00,000 and it is to be supplied with water from a reservoir 6.4km away. Water is to be supplied at the rate of 140 litres per head per day and half the supply is to be delivered in 8hrs. The full supply level of the reservoir is R.L. 180.00m and its lower water level is R.L. 105.00m. The delivery end of the main is at R.L. 22.50m and the head required for these is 12m. Find the diameter of the pipe. Take f = 0.04. 7M

(OR)

10. (a). Explain the equivalent pipe. Give the Dupuit's equation. 7M

(b). Write a short note on any two of the following; 7M

(i) Siphon
(ii) Transmission of power through pipes
(iii) Flow through nozzle at the end of pipe
(iv) Water hammer in pipes

[B17 CE 2202]

[B17 CE 2203]

II B. Tech II Semester (R 17) Regular Examinations

ENVIRONEMNTAL ENGINEERING - II

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a). Discuss different sewerage systems along with their relative merits and demerits 7M
   (b). Explain drop manhole and inverted siphon with neat sketches 7M

(OR)

2. (a). Explain briefly the method of finding the size of sanitary sewer if the discharge to be carried through them is known 7M
   (b). What factors govern the selection of sewer material and discuss the safety measures required for sewer workers 7M

UNIT-II

3. (a). Explain house plumbing systems with neat sketches 7M
   (b). Explain functions of gully trap and intercepting trap with neat sketches 7M

(OR)

4. (a). Explain the working of pumping station in detail with a neat sketch 7M
   (b). Explain the factors affecting the design of a storm water sewer 7M

UNIT-III

5. (a). What is BOD? Explain the procedure for estimation of 5 day BOD 7M
   (b). Explain the working of Grit Chamber with a neat sketch 7M

(OR)

6. (a). What is sedimentation tank? Explain its working principle and operation in detail. 7M
   (b). What is meant by biological decomposition of sewage? Explain different types of decomposition in detail 7M

UNIT-IV

7. (a). Explain the working of trickling filter with a neat sketch 7M
   (b). What is meant by activated sludge process? Explain the properties of activated sludge and its action 7M

(OR)

8. (a). Design an oxidation pond for treating sewage from a hot climatic residential colony with 5000 persons, contributing sewage @120 lpcd. The 5 – day BOD of sewage is 300 ppm 7M
   (b). Determine the size of a high rate trickling filter for the following data: 7M
   Flow = 4.5 Mld
   Recirculation ratio = 1.4
   BOD of raw sewage = 250 mg/L
   BOD removed in primary clarifier = 25%
   Final effluent desired = 50 mg/L
   Calculate also the size of the standard rate trickling filter to accomplish the above requirement

UNIT-V

9.	(a).	List the objectives of sludge digestion. Explain various methods of sludge treatment processes	7M
	(b).	Explain different sewage disposal methods	7M
		(OR)	
10.	(a).	Mention the various methods of disposal of effluent from septic tank	7M
	(b).	Explain the characteristics of sludge produced from different unit operations in a sewage treatment plant	7M

[B17 CE 2203]

[B17 CE 2204]

II B. Tech II Semester (R 17) Regular Examinations

CONCRETE TECHNOLOGY

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1.	(a).	What are Bogue's compounds? Explain their significance	7M
	(b).	List out the laboratory tests for cement. Explain determination of setting times of cement.	7M
		(OR)	
2.	(a).	Explain AAR. What are the factors promoting AAR? Explain.	7M
	(b).	Explain bulking of fine aggregate. How do you determine it?	7M

UNIT-II

3.	(a).	What is Workability & explain various factors influencing the Workability.	7M
	(b).	Explain various steps in manufacture of concrete?	7M
		(OR)	
4.	(a).	Explain the effect of time and temperature on workability	7M
	(b).	Explain in brief Segregation and Bleeding	7M

UNIT-III

5.	(a).	With neat diagram of the testing equipment describe the procedure for evaluation of compressive strength of concrete.	7M
	(b).	Explain effect of Height to diameter ratio of cylinder on strength of concrete.	7M
		(OR)	
6.	(a).	What is the importance of Non-Destructive tests? Why various Non-destructive methods of testing concrete have been developed	7M
	(b).	Explain in brief the tests on composition of Hardened concrete	7M

UNIT-IV

7.	(a).	Explain the elastic properties of concrete.	7M
	(b).	Explain the shrinkage of concrete. How it is classified? Explain any one of them.	7M
		(OR)	
8.	(a).	Explain phenomenon of creep in concrete. Explain measurement of creep with loading diagram.	7M
	(b).	Explain various factors influencing creep in concrete	7M

UNIT-V

9.	(a).	Explain what is mix design and its practical necessity.	7M
	(b).	What are different variables in proportioning that influence mix design	7M
		(OR)	
10.	(a).	Explain about High Performance Concrete.	7M
	(b).	Explain about fiber reinforced concrete.	7M

[B17 CE 2204]

[B17 CE 2205]

II B. Tech II Semester (R 17) Regular Examinations

SURVEYING II

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a). Explain the permanent adjustments of a transit theodolite 7M
   (b). Explain the methods of repetition and reiteration for measuring horizontal angle using theodolite 7M

(OR)

2. (a). Explain in detail the different errors in theodolite surveying. 7M
   (b). Explain in detail the procedure for measuring a magnetic bearing and deflection angle. 7M

UNIT-II

3. (a). Discuss the common methods of adjusting traverse. 7M
   (b). Compute the length CD for the following traverse of A, D, E points are to lie on a straight line 7M

Line	Bearing	Length (m)
AB	85°	90
BC	32°	150
CD	350°	-
DE	18°	182.0

(OR)

4. (a). Discuss the importance of Gales traverse table. 7M
   (b). Explain different methods of traversing. 7M

UNIT-III

5. (a). What are the tacheometric constants? How do you find out these constants? 7M
   (b). A tacheometer is placed at a station A and the readings on a staff held upon a BM of RL 100 m and station B are 0.640, 2.20, 3.76, 0.010, 2.120 and 4.230 respectively. The angle of depression of telescope in the first case is - 7°42'. Find the horizontal distance from A to B and the R.L of station B. Constants are 100 and 0.3. 7M

(OR)

6. (a). Discuss tangential method of tacheometry with any two cases. 7M
   (b). The horizontal angle subtended in a theodolite by a subtense bar with targets 3m apart is 18'. Compute the horizontal distance between instrument and the bar. Deduct the error of horizontal distance if the bar was 1° from being normal to the line joining the instrument and the bar station. 7M

UNIT-IV

7. (a). Two tangents intersect at a chainage of 1500 m, the deflection angle being 40°. Calculate all the data necessary for setting out a curve with a radius of 350 m by deflection angle method. 7M
   (b). Under what circumstances reverse curves are provided and what are the disadvantages of a reverse curve? 7M

(OR)

8.	(a).	What is a transition curve? What are the advantages of a transition curve?	7M
	(b).	Explain the procedure of setting out a curve using two theodolite method.	7M

UNIT-V

9.	(a).	Explain the principles of GPS and GIS in detail.	7M
	(b).	Explain the applications of Remote sensing in civil engineering.	7M

(OR)

10.	(a).	What is EDM? Explain its features and applications.	7M
	(b).	Explain photogrammetric surveying in detail.	7M

[B17 CE 2205]

[B17 CE 2206]

II B. Tech II Semester (R 17) Regular Examinations

REMOTE SENSING AND GIS

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1.	(a).	Define Remote Sensing. Illustrate the basic components of Remote Sensing.	7M
	(b).	Explain electromagnetic radiation interaction with atmosphere.	7M
		(OR)	
2.	(a).	What are the advantages and disadvantages of various Remote Sensing platforms.	7M
	(b).	What is Sensor? Write a note on classification of Sensors.	7M
		UNIT-II	
3.	(a).	Evaluate the elements of visual interpretation in image analysis.	7M
	(b).	Compare supervised and unsupervised classification.	7M
		(OR)	
4.	(a).	Evaluate the digital image processing techniques.	7M
	(b).	Explain in detail about the concept of resolution, discuss in brief about spatial and radiometric resolutions.	7M
		UNIT-III	
5.	(a).	Expand GIS. Describe the key components of GIS in detail with a neat sketch.	7M
	(b).	Distinguish the raster data models and vector data models.	7M
		(OR)	
6.	(a).	Explain the role of Map Projections in GIS.	7M
	(b).	Briefly explain the application areas of GIS.	7M
		UNIT-IV	
7.	(a).	Demonstrate the role of Remote Sensing in forest studies.	7M
	(b).	Explain the significance of Remote Sensing and GIS in Urban Planning.	7M
		(OR)	
8.	(a).	Explain the function of Remote Sensing in Geological studies.	7M
	(b).	Demonstrate the role of Remote Sensing in Land Use Land Cover studies.	7M
		UNIT-V	
9.	(a).	Explain the role of Remote Sensing for Watershed Management.	7M
	(b).	Illustrate the role of RS & GIS in Flood zoning and its mapping.	7M
		(OR)	
10.	(a).	Elucidate the role of GIS for Environmental Impact Assessment.	7M
	(b).	GIS for Ground Water prospects mapping - Explain.	7M

[B17 CE 2206]

[B17 CE 3101]

III B. Tech I Semester (R17) Regular Examinations

STRUCTURAL ANALYSIS – II

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. Using the unit load method, analyse the plain truss shown in the fig.01 and find the forces in the members. [14 M]

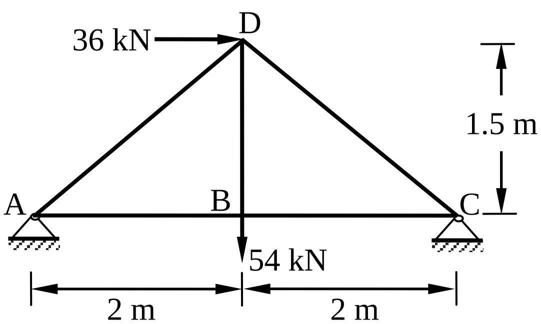


Fig. 01

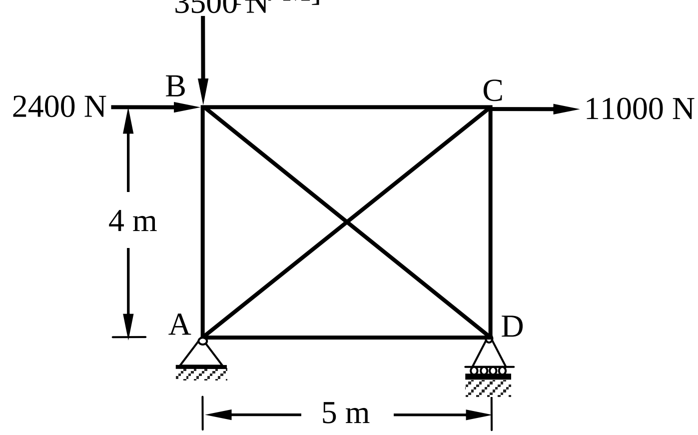


Fig. 02

(OR)

2. Find the forces in the members of the given truss shown in fig.02. Cross section area of vertical members is 28 cm^2 and for the other members is 20 cm^2. Take E = 2 × 10^5 M Pa. Use castigliano's theorem II. [14 M]

UNIT-II

3. Analyse the portal frame shown in fig.03 by slope deflection method. Draw shear force diagram. [14 M]

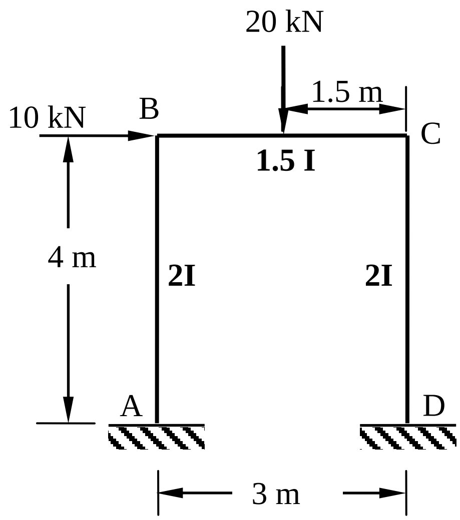


Fig.03

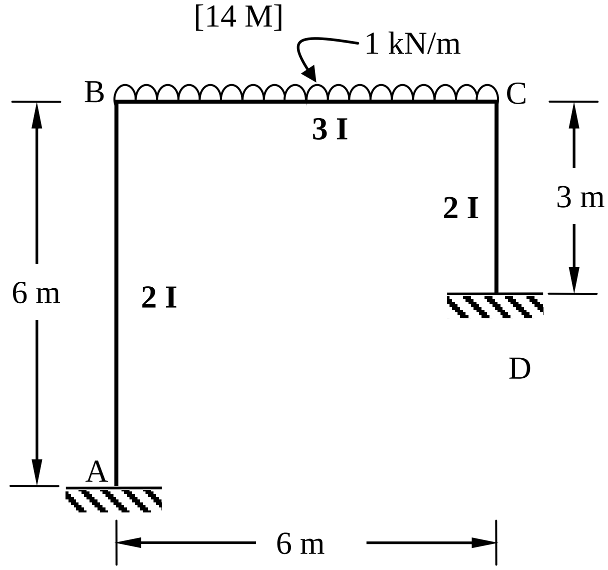


Fig. 04

(OR)

4. Analyse the portal frame shown in fig.04 by rotation contribution method(Kani's method). Sketch bending moment diagram. [14 M]

UNIT-III

5. A symmetrical three hinged parabolic arch of span 40 m and rise 8 m carries a point load of 30 kN at 10 m horizontally from the left hand hinge. The hinges are provided at the supports and at the center of the arch. Calculate the reactions at the supports also calculate the bending moment, radial shear and normal thrust at a distance of 10 m from the left support. Also calculate the maximum Positive and B.M and maximum Negative B.M. [14 M]

(OR)

6. A two hinged parabolic arch has span 40 m and rise of 6 m it has second moment of arch varies as secant of the slope of rib axis and carries uniformly distributed load 30 kN/m over left half of the spam together with concentrated load of 120 kN act at 5m from right support. Calculate the reactions and horizontal thrust at the ends and point out the values of maximum positive and negative moments and also find out the radial shear and normal thrust at 10m from right support [14 M]

UNIT-IV

7. A suspension cable hangs between two points A and B, separated horizontally by 100 m. The position of B is 16 m above A. The lowest point in the cable is 3 m below A. The stiffening girder weight 2.5 kN/m and is hinged vertically below the point A,B and the lowest point of the cable . Calculate the maximum tension which occurs in the cable when a 200 kN load crosses the girder from A to B. [14 M]

(OR)

8. A suspension bridge of 120 m span has two three hinged stiffening girders supported by two cables having a central dip of 12 m. the road way has a width of 6 m. The dead load on the bridge is 5 kN/m^2 while the live load is 10 kN/m^2 which acts on the left half of the span. Determine the shear force and bending moment in the girder at 30 m from the left end. Find also the maximum tension in the cable of position of live load. [14 M]

UNIT-V

9. Analyse the two span continuous beam shown in the fig. 05 by using stiffness matrix method. [14 M]

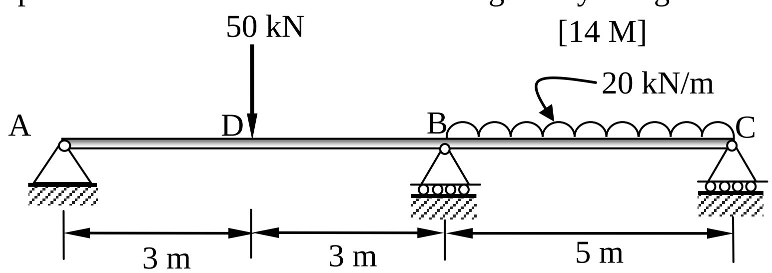


Fig. 05

(OR)

10. Analyse the two span continuous beam shown in the fig. 06 by using flexibility matrix method. Also draw bending moment diagram. [14 M]

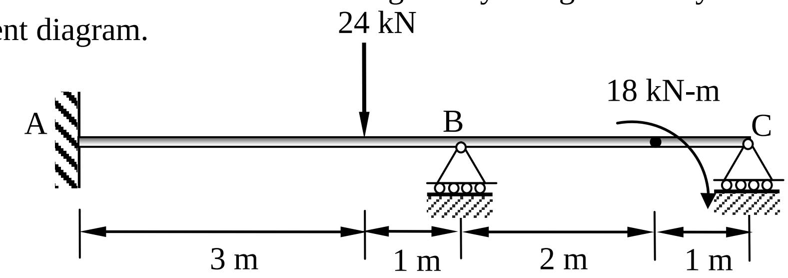


Fig. 06

[B17 CE 3101]

[B17 CE 3102]

III B. Tech I Semester (R17) Regular Examinations

REINFORCED CONCRETE STRUCTURES I

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1.	(a).	Discuss about characteristic load and design values for materials and loads.	4M
	(b).	Analyze an isolated T-beam, having a span of 6.0 m and cross section having flange width of 1300 mm, flange thickness of 100 mm, web width of 325 mm and an effective depth of 420 mm, is reinforced with 7 Nos of 28 mm dia bars on tension side. The materials used are concrete mix of grade M 20 and HYSD steel of grade Fe415.	10M

(OR)

2.	(a).	Under what circumstances are doubly reinforced beam used and what are the advantages of doubly reinforced beams over singly reinforced beams	4M
	(b).	A rectangular beam of 7m span (centre-to-centre of supports) resting on 300 mm wide simple supports, is to carry a uniformly distributed dead load (excluding self-weight) of 15 kN/m and a live load of 20 kN/m. Using Fe 415 steel and beam subjected moderate exposure condition, design the beam section at mid span.	10M

UNIT-II

3.	(a).	Stirrups may be vertical or inclined. When does it become mandatory to use vertical stirrups.	2M
	(b).	A simply supported beam 300 × 600 mm (effective) is reinforced with five bars of 25 mm diameter. It carries a characteristic load of 80 kN/m(including its own weight) over an effective span of 6 m. out of five main bars, two bars can be bent-up safely near the support. Design the shear reinforcement for the beam. Using M 20 grade of concrete and Fe 415 steel.	12M

(OR)

4.	(a).	Briefly discuss the different modes of failure under combined flexure and torsion	7M
	(b).	Design a rectangular beam section, 300 mm wide and 550 mm deep(overall), subjected to an ultimate twisting moment of 25 kNm, combined with an ultimate bending moment of 60 kNm and an ultimate shear force of 50 kN. Assume M 20 concrete and Fe 415 steel.	7M

UNIT-III

5.	(a).	Explain the difference in the behavior of one way slabs and two way slabs	7M
	(b).	A hall in a building has a floor consisting of a one-way continuous slab cast monolithically with simply supported 250 mm wide beams spaced at 4 m centre-to-centre. The clear span of the beam is 9 m. Assuming a live load on the slab as 3.0 kN/m^2, partition load as 1 kN/m^2 , and load due to finishes as 0.6	7M

kN/m^2. Design the slab with M 20 grade concrete and Fe 415 steel.

(OR)

6. (a). Explain the need for corner reinforcement in two-way rectangular slabs whose corners are prevented from lifting up. 2M

(b). Design a simply supported slab to cover a hall with internal dimensions 4.0 m × 6.0 m. The slab is supported on masonry walls 230 mm thick. Assume a live load of 3 kN/m^2 and a finish load of 1.0 kN/m^2. Use M 20 concrete and Fe 415 steel. Assume that the slab corners are prevented from lifting up. 12M

UNIT-IV

7. (a). Distinguish between (i) unsupported length and effective length of a compression member; (ii) braced column and unbraced column. 2M

(b). Design the reinforcement for a column with l_{ex} = l_{ey} = 3.5 m and size 300 × 500 mm, subject to a factored axial load of 1250 kN with biaxial moments of 180 kNm, and 100 kNm(i.e., M_{ux} = 180 kNm, M_{uy} = 100 kNm). Assume M 25 concrete and Fe 415 steel. 12M

(OR)

8. (a). Enumerate the functions of the transverse reinforcement in a reinforced concrete columns. 2M

(b). Design a short square column, with effective length 3.0 m capable of safety resisting the following factored load effects (under uniaxial eccentricity) $P_u = 1625\ kN, M_u = 75\ kNm$ Assume M 25 concrete and Fe 415 steel. 12M

UNIT-V

9. (a). What are the types of shear checks to be done in isolated footings? Explain. 2M

(b). Design an isolated footing for a rectangular column 300 × 500 mm, reinforced with 6-25 mm ø bars, and carrying a factored axial load of 1000 kN and afactored uniaxial moment $M_{ux} = 120\ kN/m$ (with respective to the major axis) at the column base. Assume the moment is reversible. Assume soil with an allowable pressure of 200 kN/m^2 at a depth of 1.25 m bellow ground. Assume Fe 415 grade steelfor column and footing, and M 20 grade concrete for the footing and M 25 grade concrete for column. 12M

(OR)

10. (a). Explain the basic difference in structural behaviour between 'stair slabs spanning transversely' and stair slabs spanning longitudinally 7M

(b). Design a dog-legged staircase('waist slab') for an office building, assume a floor-to-floor height of 3.0 m, a flight width of 1.2 m, and a landing width of 1.25 m. Assume the stairs to be supported on 230 mm thick masonary wall at the edges of the landing, parallel to the risers. Assume live loads of 5.0 kN/m^{2}and mild exposure conditions. Use Fe 415 steel. 12M

[B17 CE 3102]

[B17 CE 3103]

III B. Tech I Semester (R17) Regular Examinations

STEEL STRUCTURES I

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a). State the advantages of welded connections over bolted connections? 2M

   (b). Determine the safe load and efficiency of a single cover butt joint. The main plates are 18 mm thick connected by 22 mm diameter bolts at a pitch of 150 mm. Design the cover plate also. What is the percentage reduction in the efficiency of the joint if the plates are lap jointed. 12M

(OR)

2. (a). Why are drilled holes preferable than punched holes? 2M

   (b). Design a butt joint to connect two plates 200 ×8 mm (Fe 410 grade) using M 16 bolts. Arrange the bolts to give maximum efficiency. 12M

UNIT-II

3. (a). Why are slot and plug welds not preferred. 2M

   (b). Design a connection to joint two plates of sizes 250 mm ×12 mm of grade Fe 410 to mobilize full plate tensile strength using shop fillet welds if (i) a lap joint is used (ii) a double cover butt joint is used. 12M

(OR)

4. (a). Under what circumstances are slot and plug welds used. 2M

   (b). A tie member of truss consisting of an angle section ISA 75×75×8 mm of Fe 410 grade, is welded to a 10 mm gusset plate. Design a weld to transmit a load equal to the ful strength of the member. Assume shop welding. 12M

UNIT-III

5. (a). What is lug angle? Why lug angles are used. 4M

   (b). Design a tension member to transmit a pull of 150kN.Effective length is 4.5m.Member should consist of a pair of angles connected to the one side of the gusset plate of thickness 10mm. Use 20 mm bolts of grade 4.6 10M

(OR)

6. (a). What is meant by shear lag. 2M

   (b). A diagonal member of a roof carries an axial tension of 500 kN. Design the section and its connection with a gusset plate. Use $f_y = 250\ MPa, f_u = 410\ MPa$ 12M

UNIT-IV

7. (a). What is the main purpose of lacings and battens. 7M

   (b). Design a column with two channels connected back to back and battened to support a load of3500kN.The effective length of the column is 5m. Design battens also. 7M

(OR)

8. Design a built up column with two channels placed face-to-face and separated apart. The column is of 6.6 m effective length and supports a factored load of 1500 kN. Also design the lacing system. 14M

UNIT-V

9. Design a laterally supported beam of effective span 6 m for the following data. Grade of steel Fe 410, maximum bending moment M = 150 kNm, maximum shear force V = 210 kN, check for deflection is not required. 14M

(OR)

10. (a). Distinguish between laterally supported and unsupported beams? 4M

(b). Design a laterally unsupported beam with 6m simply supported effective span,subjected to a point load of 40 kN at mid span, select a section of ISMB 300 mm. 10M

[B17 CE 3103]

[B17CE3104]

III B. Tech I Semester (R17) Regular Examinations

GEOTECHNICAL ENGINEERING-I

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a). Derive the Relation $\rho_{d} = \frac{(1-n_a)G.\rho_w}{1+WG}$ 7M

(b). In a field exploration, a soil sample was collected in a sampling tube of internal diameter 5cm below ground water table. The length of the extracted sample was 10.2cm and its weight was 387gm.If G=2.7,and the weight of the dried sample is 313gm,Determine the porosity, Void Ratio, Degree of Saturation, Bulk Density and dry density of the sample. 7M

(OR)

2. (a). Discuss about Indian Standard Classification System 7M

(b). A soil sample obtained from a core cutter is 500mm in diameter and 100mm in height. Its wet unit weight is $17kN/m^3$ and dry unit weight is $13kN/m^3$. If the specific gravity of solids is 2.8 determine the volume of solids of the specimen and also determine the moisture content in the soil specimen. 7M

UNIT-II

3. (a). A Sand deposit consists of two layers. The top layer is 2.5mt thick(ρ=1709.67Kg/m^3) and the bottom layer is 3.5m thick (ρ_{sat}=2064.52Kg/m^3).The water table is at a depth of 3.5m from the surface and the Zone of capillary saturation is 1 m above the water table ,Determine the Total, Neutral and effective stresses. 7M

(b). Explain the factors Effecting Permeability 7M

(OR)

4. (a). Explain Quick sand condition and critical hydraulic gradient. 7M

(b). A Constant head Permeability test was carried out on a cylindrical sample of sand 10cms dia& 15cm height 160cm3 of water was collected in 1.75 mins. Under a head of 30cm. Compute the Coefficient of permeability (K) & Velocity of Flow. 7M

UNIT-III

5. (a). Derive an expression for the vertical stress at a point due to a point load, using Boussinesq's theory. 7M

(b). A square foundation 5m x 5m is to carry a load of 4000kN. Calculate the vertical stress at a depth of 5m below the centre of the foundation I_N=0.084 for m=n=0.5. Also determine the vertical stress using 2:1 distribution. 7M

(OR)

6. (a). Explain about New mark's Influence Chart 7M

(b). A Concentrated Load of 2000kNis applied at the ground surface. Determine the vertical stress at a point P which is 6m directly below the load. Also calculate the vertical stress at a point R which is at a depth of 6m but at a horizontal distance of 5m from the axis of the load. 7M

UNIT-IV

7. (a). What is the Effect of Compaction on Engineering Properties of Soil? Discuss in briefly 7M

(b). A Stratum of Clay is 2m thick and has an initial overburden pressure of $50kN/m^2$ at its middle. Determine the final settlement due to an increase of $40kN/m^2$ at the middle of the clay layer. The clay is over-consolidated, with a pre consolidation pressure of $75kN/m^2$.The values of the coefficients of recompression and Compression index are 0.05 and 0.25, respectively. Take initial Void ratio as 1.40 7M

(OR)

8. (a). Discuss Terzaghi's theory of consolidation, Stating various assumptions 7M

(b). A 3m thick clay layer beneath a building is over lain by a permeable stratum and i s under lain by an impervious rock. The coefficient of consolidation of the clay was found to be 0.025cm2/minute. The final expected settlement for the layer is 8cm, (a) How much time will it take for 80% of the total settlement to take place? (b)Determine the time required for a settlement of 2.5cm to occur (c)Compute the settlement that would occur in one year. 7M

UNIT-V

9. (a). What are the advantages of triaxial shear test over the direct shear test 7M

(b). In a direct shear test, the following results are obtained. 7M

Normal stress (kN/m^2)	25	50	75
Shear stress at failure (kN/m^2)	30	45	60

Size of soil sample is 60mm x 60mm

Find the shear parameters if a triaxial test was conducted on the same soil with a cell pressure of $30kN/m^2$, what would be the deviator stress at failure?

(OR)

10. (a). How the shear tests classified is based on drainage conditions? What practical situations do they simulate? 7M

(b). A shear Vane of 7.5cm diameter and 11.00cm length was used to measure the shear strength of soft clay. If a torque of 600N-m was required to shear the soil, Calculate the shear strength. The Vane was then rotated rapidly to cause remolding of the soil. The Torque Required in the remoulded state was 200N-m.Determine the Sensitivity of the soil. 7M

[B17CE3104]

[B17CE3105]

III B. Tech II Semester (R17) Regular Examinations

FLUID MECHANICS -II

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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UNIT-I

1. (a). Derive an expression for drag force on a smooth sphere of diameter of D moving with a Uniform velocity V in a fluid of density ρ having Dynamic viscosity μ using Rayleigh's Method. 7M

   (b). Explain Boundary Layer separation with a neat sketch 7M

(OR)

2. (a). Explain different model laws used in Dimensional Analysis along with examples 7M

   (b). Air flows over a flat 1m long at a velocity of 6 m/s. Determine 7M

   The boundary layer thickness at the end of the plate

   Shear stress at the middle of the plate

   The total drag per unit length on the sides of the plate

   Take density of air ρ = 1.226 kg/m3 and kinematic viscosity υ = 0.15 x 10-4 m2/ s

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UNIT-II

3. (a). Derive an expression for force exerted by jet and work done by the jet on a movingvertical flat plate with a neat sketch 7M

   (b). A kite weighing 12.26 N has an effective area of 0.9 m^2 The tension in the kite string is 32.37N when the string makes an angle of 45^O with the horizontal. For a wind of 32 km/hr,what are the coefficients of lift and drag if the kite assumes an angle of $8^O$with the horizontal. Take specific gravity of air as 11.801 kg/m^3 7M

(OR)

4. (a). Explain different types of Drag and Magnus effect with neat sketches 7M

   (b). A jet of water of Diameter 7.5cm strikes a curved plate at its centre with a velocity of 20 m/s. The curved plate is moving with a velocity of 8 m/s in the direction of the jet. The jet is deflected through an angle of 165^O. Assume the plate is smooth find 7M

   exerted on the plate in the direction of the jet

   Power of the jet

   Efficiency of the jet

UNIT-III

5. (a). Explain (1)cavitation, (2)Impulse Turbine and(3) Draft tube provideexamples where ever necessary — 7M

(b). A pelton wheel has a mean bucket speed of 12 m/s and is supplied with water at arate of 750 litres per second under a head of 35 m. If the bucket deflects the jet through an angle of 160^{O}, find the power developed by the turbine and its hydraulic efficiency.Take the coefficient of velocity as 0.98. Neglect friction in the bucket. Also determinethe overall efficiency of the turbine if its mechanical efficiency is 80% — 7M

(OR)

6. (a). Define Hydraulic efficiency, Mechanical efficiency, Volumetric efficiency, and overall Efficiency — 7M

(b). A Pelton wheel developes 5520 kW under a head of 240m at an overall efficiency of 80% when revolving at a speed of 200 r.p.m. Find the unit discharge, unit power and unit speed. — 7M

UNIT-IV

7. (a). Define Net positive suction head ,specific speed of centrifugal pump and operating characteristic curves — 7M

(b). Determine the maximum speed at which a double acting reciprocating pump can be operated under the following conditions (1) no air vessel on the suction side (2) a very large air vessel on the suction side close to the pump. The suction lift is 4m, length of the suction pipe is 6.5m, diameter of suction pipe is 100mm, diameter of piston is 150mm and length of stroke is 0.45m.Assume simple harmonic motion, atmospheric head as 10.3m of water and separation occurs at 2.6m of water absolute.
Take Darcy's f = 0.024 — 7M

(OR)

8. (a). Find the power required to drive a centrifugal pump which delivers 40liters of water per second to height of 20m through a 150mm diameter and 100 mm long pipeline. The overall efficiency of pump is 70% and Darcy's f= 0.06 for the pipeline. Assume the suction pipe inlet losses is equal to 0.33 m — 7M

(b). Explain — 7M
(1) Types of reciprocating pumps (2) Slip of reciprocating pump (3) Air vessels With neat sketches where ever required.

UNIT-V

9. (a). Derive expressions for Hydraulic mean radius and the side slope of the economical trapezoidal channel. — 7M

(b). Discharge of 10 m^3/s is allowed to pass through a rectangular channel of width 3.5mhaving a depth of water equal to 1.5 calculate critical depth of that channel and the corresponding specific energy. — 7M

(OR)

10. (a). A discharge of 25m^3/s is allowed in a trapezoidal channel having a base width of 2.5m and side slopes equal to 1:1.5 depth of water in the channel is equal — 7M

1.5m and longitudinal slope of the channel is equal to 1 in 5000 value of manning's coefficient is equal to 0.015. Calculate the velocity of flow and Chezy's constant C

(b). During a rapidly varied flow the post jump depth is equal 2.5m and the Froude number corresponding to post jump is equal to 0.5 calculate the pre jump depth, energy loss in the rapidly varied flow and height of the jump. 7M

[B17CE3105]

[B17CE3106]

III B. Tech II Semester (R17) Regular Examinations

ESTIMATION AND QUANTITY SURVEYING

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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UNIT-I

1. (a). State and explain different types of Sanctions/Approvals 7M

(b). Explain the following: 7M

i) SoR

ii) Work Charged Establishments

iii) Quantity surveying

(OR)

2. (a). Explain the errors in estimation? 7M

(b). State and explain different types of estimates 7M

UNIT-II

3. Estimate the quantities of the building items of a two roomed building from the given plan and section as shown in figure:1 14M

Lime concrete in foundation

1st class brick work with 1:6 cement mortar in foundation and plinth

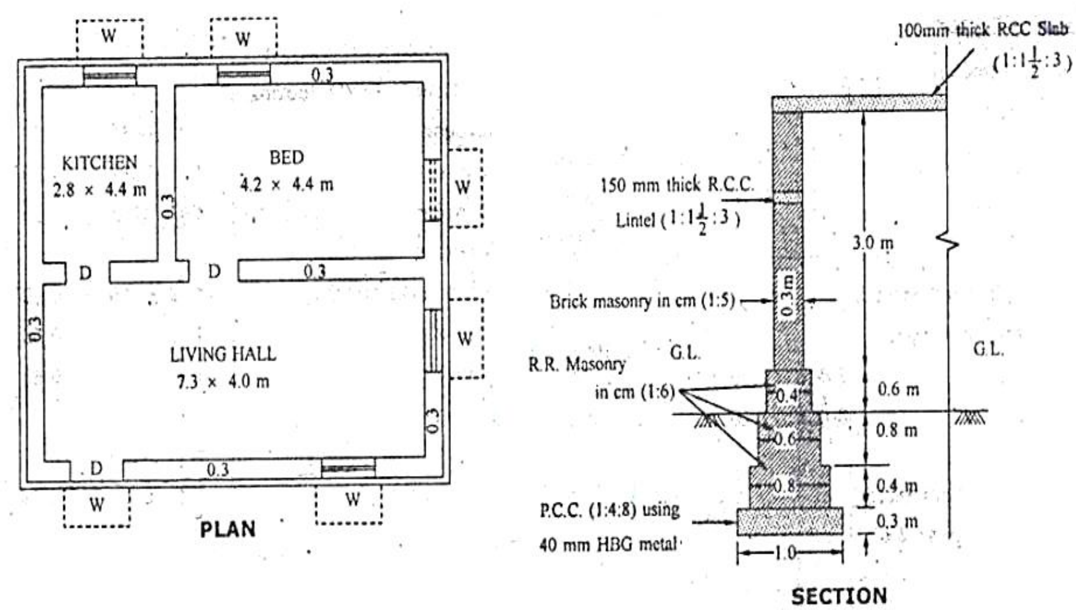


Figure:1

(OR)

4. (a). Prepare a detailed estimate of the following items for a R.C.C. beam of 8 meter clear span and 75cm X 40cm in section from the given figure:2 14M

R.C.C. work including centering and shuttering

Steel in detail shall be calculated separately and

Prepare a schedule of bars.

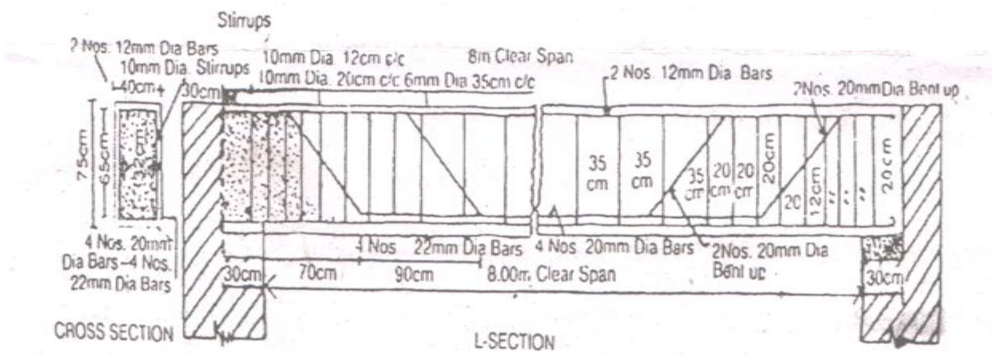


figure:2

UNIT-III

5. Prepare the rate analysis for the following items of work. Assume suitable rates for material and labour. 14M

Brick masonry work in CM (1:6)

Plastering in CM (1:4), 12mm thick for walls

(OR)

6. Give the detailed specifications for the following items of work 14M

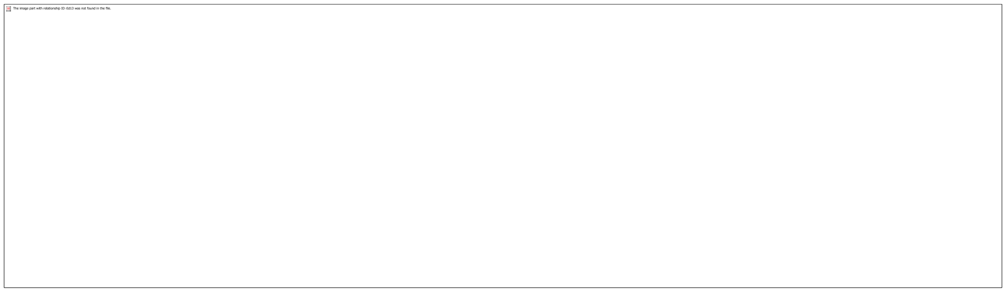
RCC (1:2:4) for roof slabs

Wood work for door and window frames

Random Rubble stone masonry in (1:3) lime sand mortar

UNIT-IV

7. Estimate the cost of earthwork for a portion of road for 360m length from the following data: 14M



The formation width of the road is 10m and the side slopes are 2:1 in banking and 1.5:1 in cutting

(OR)

8. The ground levels along the center line of the road are given below 14M

Chainage (meters)	0	50	100	150
RL of Ground	97.00	96.50	96.00	97.50

The road is to be formed in embankment with the formation level at 100.00m throughout the length. If the road width is10.00 m and the side slopes 2:1, calculate the quantity of earthwork required by Trapezoidal rule. Assume transverse slope as level

UNIT-V

9. (a). Define valuation. Explain briefly the valuation methods. 7M

(b). An old building has been purchased by a person @ a cost of Rs. 6,00,000/- excluding the cost of land. Calculate the amount of annual sinking fund @ 9% interest assuming the life of building as 30 years and the scrap value of the building as 10% of the purchase. 7M

(OR)

10. A freehold property having an area of 800 sq.m. is jointly held by four brothers and it is fully developed. It’s a 4 storeyed building having a basement. The structure is being designed and used as a college building. It is an RCC framed structure given for a monthly rent of Rs. 8000/- (for whole building). The usual outgoings may be taken as 20% of gross annual rent. Work out the share of each brother in property. 14M

[B17CE3106]

[B17 CE 3201]

III B. Tech II Semester (R17) Regular Examinations

REINFORCED CONCRETE STRUCTURES II

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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UNIT-I

1.	(a).	What is the purpose of a retaining wall? What are the different types of concrete retaining walls?	2M
	(b).	A cantilever-retaining is required to retain earth 3.8 m high above the ground level. The backfill surface is inclined at an angle 15º with the horizontal and the backfilled soil has a unit weight of 18 kN/m^3 and an angle of internal friction of 30º. The exposure condition is moderate. Assume that the SBC of soil is 150 kN/m^2 and that the coefficient of friction between the soil and concrete is 0.5. Design the RC retaining wall.	12M

(OR)

2.	(a).	What is the purpose of a shear key? Describe its action.	2M
	(b).	Design a counterfort-type retaining wall to retain a 6.8 m high backfill above the ground lavel. The unit weight and SBC of soil at the site are 18 kN/m^{3}and 170 kN/m^2,respectively. The angle of internal friction of soil and coefficient of friction are 30º and 0.6 respectively. The exposure condition is moderate.	12M

UNIT-II

3.	(a).	What are the types of joints used in water tanks.	2M
	(b).	Design an R.C tank of internal dimensions 12 m × 3.5 m × 3.5 m. The tank is to be provided underground. The soil surrounding the tank is get wet. Adopt suitable working stress. Soil weighs 1800 kg/ cubic metre. Use M 20 concrete and Fe 250 steel.	12M

(OR)

4.	(a).	An open square tank is 5 m × 5 m × 3 m deep and supported 6 m above the ground level on beam on beams and columns. Design the tank and the supporting beams columns and foundations. Use M 20 concrete and Fe 250 steel. Sketch the reinforcement details of water tank.	14M

UNIT-III

5.	(a).	Define economical span	2M
	(b).	Design a RC slab bridge of span 4.5 m for IRC class AA (tracked vehicle) with M 25 concrete and Fe 415 steel. Assume width of roadway as 7.5 m with footpath of 600 mm on either side. Draw the cross sectional details of the slab.	10M

(OR)

6.		Design a RCC Tee beam girder bridge to suit the following data Clear width of roadway = 7.5 m Span (centre to centre of bearing)= 16 m	14M

Live load: I.R.C class AA tracked vehicle
Average thickness of wearing coat= 80 mm
Use M 25 grade and Fe 415 steel, using courbon's method, compute the design moments and shear, design of deck slab, main girder and cross girders and sketch the typical details of reinforcement.

UNIT-IV

7. (a). Under what circumstances are pile foundations preferred. 2M
(b). An RC column of size 400×400 mm is supported on four piles of 300 mm diameter (bored cast in situ piles). The column carries a load of 750 kN and a moment of 250 kNm in the $x - x$ direction. Design the pile cap assuming M 25 concrete and Fe 415 steel. Further, assume that the piles are capable of resisting the reaction from the pile cap. 12M

(OR)

8. (a). Design precast pile of diameter 400 mm carrying an axial load of 275 kN placed in submerged medium dense sandy soil having an angle of internal friction of 32^o. The density of soil is 18 kN/m^3 and submerged density of soil is 10 kN/m^3. Angle of wall friction between concrete pile and soil, δ is 0.75 $\emptyset$= 24^o. Assume the following data: Depth of top of pile cap below ground level is 500 mm, thick ness of pile cap is 1.5 m grade of concrete in pile is M 25, Fe 415 steel used and clear cover to reinforcement is 75 mm. Determine the vertical capacity of the pile in accordance with IS 2911 (Part I section I) and design the pile. 14M

UNIT-V

9. (a). How is the combined shear and moment transfer in flat slab consider 2M
(b). Design the interior panel of flat slab of size 5 m×5 m with suitable drop to support a live load of 4 kN/m^2. The floor is supported by columns of size 450 mm× 450 mm. Use M 20 concrete and Fe 415 steel. Sketch the reinforcement details by cross sections i) at column strip ii) at middle strip. 12M

(OR)

10. (a). Briefly explain the approximate analysis of grid floor according to IS 456:2000 2M
(b). A reinforced concrete grid floor is to be designed to cover a floor area of 12 m × 18 m. The spacing of the ribs in mutually perpendicular directions is 1.5 m center to center . Live load on floor is 3 kN/m^2. Adopt M 20 grade concrete and Fe 415 grade HYSD bars. Assume ends are simply supported. Analyse the grid floor by IS 456:2000 method and design suitable reinforcements in the gride floor. 12M

[B17 CE 3201]

[B17 CE 3202]

III B. Tech II Semester (R17) Regular Examinations

STEEL STRUCTURES II

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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UNIT-I

1.	(a).	Differentiate between a slab base and gusset base.	2M
	(b).	Design a suitable slab base for a column section ISHB 200@365.9 N/m supporting an axial load of 400 kN. The base plate is to rest on concrete pedestal of M 20 grade.	12M
		(OR)	
2.	(a).	Explain types of slab bases.	2M
	(b).	Design a gusseted base for a column ISHB 250 @64.96kg/m two cover plates 300 mm × 20mm one to each flange to transmit an axial load of 1800kN.The concrete pedestal is M 20 grade.	12M
		UNIT-II	
3.	(a).	What is the difference between a beam and plate girder.	2M
	(b).	Design a welded plate girder 24 m in span and laterally restrained throughout. It has to support a uniform load of 100 kN/m throughout exclusive of self weight. Design the girder without intermediate transverse stiffeners. Design the cross section with end load bearing stiffeners and connections.	12M
		(OR)	
4.	(a).	How does the plate girder derive post-buckling strength.	2M
	(b).	Design the plate girder using intermediate transverse stiffeners. Connections need not be designed. Use post critical method for the design. Maximum bending moment 11,448 kNm and shear force1908 kN.	12M
		UNIT-III	
5.	(a).	Write brief note on suspended bottoms in water tanks	2M
	(b).	Design an elevated cylindrical steel tank with hemispherical bottom, for a capacity of 1,20,000 liters. The tank has conical roof and its ring beam is 15 m high above the G.L. Take basic wind pressure as 1.5kN/m^2.	12M
		(OR)	
6.	(a).	For an elevated cylindrical tank of height 5 m and diameter 5 m, estimate the wind pressure on the tank, if basic wind pressure is 1.5 kn/m^2.	2M
	(b).	Design an overhead circular of riveted steel for a capacity of 60000 litres. The tank is to be supported on four columns; each of 9 m height. The tank is to be built at Bombay.	12M
		UNIT-IV	
7		Design an unstiffened welded seat connections for beam ISMB 250@365.9 N/m transmitting an end reaction of 110 kN, due to the factored loads the flange of the	14M

column ISHB 200 @ 365.9 N/m. The seat angle is welded to the column flange in workshop.

(OR)

8. (a). Distinguish between unstiffened and stiffened seat angle connections. 7M

(b). Design a stiffened welded seat connection between a beam ISMB 400 @604.3 N/m and column ISHB 200@ 365.9 N/m for a reaction of beam 120 kN. Assuming Fe 410 grade steel and site welding. 7M

UNIT-V

9. Design a rocker bearing for a bridge girder having the following data 14M

i) D.L + L.L+ I.L reaction: 1200 kN

ii) Reaction due to wind Overturning effect: 230 kN

iii) Lateral load due to wind: 80 kN

iv) Longitudinal force: 360 kN

Assume any other data not given

(OR)

10. Design a roller bearing for the plate girder having following data 14M

i) Vertical load due to DL+ LL+ Impact = 924.4 kN

ii) Vertical load due to wind overturning= $\frac{1}{2} \times 395.6 = 197.8$

Total vertical reaction= 924.4+197.8 = 1122.2 kN

iii) lateral load due to wind = 238.5 kN

iv) Longitudinal load = 294.2 kN.

[B17 CE 3202]

[B17CE3203]

III B. Tech II Semester (R17) Regular Examinations

GEOTECHNICAL ENGINEERING-II

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

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UNIT-I

1.	(a).	Explain two boring methodswith neat sketch	7M
	(b).	Describe, in brief, various geophysical methods.	7M
		(OR)	
2.	(a).	Discuss standard penetration test.Whatare the various corrections?	7M
	(b).	The measured Nvalue inaborehole is foundto be 20. Thedepthofthe borehole was 15m.Thewater tablewaslocated 1.5mbelow ground level.The soil is fine silty and its unit weight above and below water table was 16and 21kN/m3respectively. Calculatethecorrectedvalueof N.	7M

UNIT-II

3.	(a).	What are different types of shallowfoundation?Explain with the helpof sketches.	7M
	(b).	A Square footing is to be designed to carry a total load of 500KN in a soil having C=10kN/m^2,Ø=25^0,γ=16kN/m^3.The footing is to be laid at a depth of 2mts. belowground level and F.S is 3.Find the Dimension of the footing. Given N_c=22,N_q=11,N_γ=9.	7M
		(OR)	
4.	(a).	Derive the expression for Terzaghi’s Bearing Capacity theory and write the assumptions?	7M
	(b).	Afooting 4mx2minplan,transmitsapressureof150N/sq.m on a cohesive soil having E=6x10^4kN/sq.m andμ=0.50.Determine the immediate settlement of the footing at the center, assuming it to be (a) a flexible footing, I=1.32 and (b) a rigid footing,I=1.20	7M

UNIT-III

5.	(a).	Explain Pneumatic Caisson with neat sketch and mention its advantages and disadvantages.	7M
	(b).	A concrete pile, 25cm in diameter is driven in to a dense sand (Ø = 30^0, γ=20kN/m^3,K=1, Tan δ =0.70) for a depth of 6m. If the water table is at 2m from the ground level, estimate the safe load, taking Factor of safety 3.	7M
		(OR)	
6.	(a).	Discuss the Causes and Remedies for Tilts and Shifts (with neat sketches) in Well Foundations.	7M
	(b).	A pile group consists of 9 piles of 300mm dia and length 10m, is arranged in 3	7M

rows at spacing of 750mm c/c. The piles are driven into a deposit of clay whose properties are C=100kN/m^2, γ=20kN/m^3.Determine the safe load on the pile group. Take F.S=3, α =0.6,N_c=9

UNIT-IV

7. (a). Discuss the Friction circle method for the stability analysis of slopes. Can this method be used for purely cohesive soil? 7M

(b). A cut of length 10m is made in a cohesive soil deposit (C=30kN/m^2, Ø = 0^0, γ=20kN/m^3).There is a hard stratum under the cohesive soil at a depth of 12m below the original ground surface. If the required factor of safety is 1.5, determine the safe slope 7M

(OR)

8. (a). Explain types of slope failures 7M

(b). Aslopeis to be Constructionin a soil ForwhichC=0and Ø = $36^{0.}$ Itis tobeassumedthatthewaterlevelmayoccasionallyreachthesurfaceofaslope,with seepage takingplaceparalleltotheslope.Determinethemaximumslopeangleforaforafactor ofsafety1.5,assumingapotentialfailuresurfaceparalleltotheslopeWhatwouldbeth efactorofsafetyoftheslope,constructedatthisangle,ifthewatertableshouldbewellb elowthesurface? 7M

UNIT-V

9. (a). What are the assumptions of Rankine's theory? Derive the expressions for active pressure and passive pressure 7M

(b). Determine the passive pressure by Rankine's theory per unit run for a retaining wall 4m high, with i=15^0, Ø = $30^0$and γ=20kN/m^3. The back face of the wall is smooth and vertical. 7M

(OR)

10. (a). DiscussCulmann'smethodforthe determinationof active earth pressure. 7M

(b). A5mhigh rigid retainingwallhastoretainabackfillofdry,cohesionlesssoilhavingthefollowingproperties: Ø = 30^0,e=0.74andG=2.68.PlotthedistributionofRankinelateralearthpressureonthewallanddeterminethemagnitudeandpointofapplicationoftheresultantthrust. 7M

[B17CE3203]

[B17 CE 3204]

III B. Tech II Semester (R17) Regular Examinations

TRANSPORTATION ENGINEERING - I

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1. (a).	Briefly outline the highway development in India.	7M
(b).	Compare Nagpur & Bombay Road Development plans.	7M

(OR)

2. (a).	What is master plan? Explain the importance of master plan in highway planning	7M
(b).	Briefly outline about rural road development plan	7M

UNIT-II

3. (a).	Discuss the design factors of Highway alignment	7M
(b).	Calculate the safe stopping distance for design speed of 50 kmph for two-way traffic on a two lane road. Assume coefficient of friction as 0.35 and reaction time of driver as 2.5 seconds	7M

(OR)

4. (a).	Derive an expression for extra widening on horizontal curvers.	7M
(b).	Find out the length of transition curve for the following data: Radius of horizontal curve = 400m, design speed = 100 kmph, length of wheel base = 6.2m, number of lanes = 2, rainfall at the location = heavy, terrain condition = hilly, superelevation is introduced by rotating the edges with reference to center line and rate of introduction of superelevation is 1 in 150. Width of highway is 7m.	7M

UNIT-III

5. (a).	Explain how the speed and delay studies are carried out. What are the various uses of delay studies?	7M
(b).	What are the various types of parking facilities designed for traffic needs?	7M

(OR)

6. (a).	Explain the relationship between speed, travel, time, volume, density and capacity.	7M
(b).	Explain the various measures that may be taken to prevent accidents.	7M

UNIT-IV

7. (a).	Explain the test procedure of shape test for aggregates.	7M
(b).	Estimate the thickness of cement concrete pavement using the method suggested by IRC. Modulus of elasticity of concrete = 3.5 X 10^5 kg/cm^2,	7M

Modulus of rupture of concrete = 30 kg/cm^2, Poisson's ratio of concrete = 0.17, Modulus of subgrade reaction = 5 kg/cm^2, Wheel load = 4800 kg, Radius of contact area = 13 cm.

(OR)

8.	(a).	Discuss the properties of bitumen	7M
	(b).	Explain group index method of pavement design. What are the limitations of te method?	7M

UNIT-V

9.	(a).	Write a descriptive note on maintenance of highways	7M
	(b).	Write a short note on alligator and reflection cracking.	7M

(OR)

10.	(a).	What are the requirements of material, plants & equipment's for bituminous pavement construction? Discuss briefly.	7M
	(b).	Give a descriptive note on Pavement Evaluation Techniques	7M

[B17 CE 3204]

[B17CE3205]

III B. Tech IInd Semester (R17) Regular Examinations

AIR POLLUTION AND CONTROL

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1.	(a).	Define air pollution. Explain the various factors influencing air pollution.	7M
	(b).	Discuss about the typical gaseous pollutants and their sources.	7M
		(OR)	
2.	(a).	Write critical note on air quality standards.	7M
	(b).	What are the major harmful effects of particulate matter on human health and materials?	7M
		UNIT-II	
3.	(a).	Explain the plume behaviour in detail.	7M
	(b).	What is maximum mixing depth? Explain its practical significance in Air pollution control.	7M
		(OR)	
4.	(a).	What is atmospheric stability? Explain the temperature inversions with reference to air pollution.	7M
	(b).	What is meant by effective stack height? Mention guidelines for minimum.	7M
		UNIT-III	
5.	(a).	List the recorded major air pollution episodes chronologically. State their significant features.	7M
	(b).	How SOx and NOx emissions affect human health and the growth of plants?	7M
		(OR)	
6.	(a).	Explain the effects of carbon monoxide and sulphur oxides on human health.	7M
	(b).	Explain any two air pollution disasters occurred during 20th and 21st centuries.	7M
		UNIT-IV	
7.	(a).	List the sampling devices used for air quality monitoring. Explain in brief various sampling methods of gaseous pollutants.	7M
	(b).	Explain the importance of stack monitoring. List out various methods of stack monitoring.	7M
		(OR)	
8.	(a).	Write short notes on Ambient Air Quality monitoring.	7M
	(b).	What is the use of high volume sampler? Explain.	7M
		UNIT-V	
9.	(a).	Explain with the help of a neat sketch the working of an electrostatic precipitator.	7M
	(b).	Explain with the help of a neat sketch the working of a centrifugal scrubber.	7M

(OR)

10. (a). With a neat sketch, explain working of a cyclone separator. 7M

(b). When do you recommend absorption as a method of control of gaseous contaminants? 7M

[B17CE3205]

[B17CS3213]

III B. Tech II Semester (R17) Regular Examinations

DATABASE MANAGEMENT SYSTEMS

(Common to CE & ME)

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks:70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

*****

UNIT-I

1.	(a).	Who are the different database users? Explain their interfaces to database management system.	7M
	(b).	Describe the client server architecture for the database with necessary diagram.	7M

(OR)

2.	(a).	Briefly explain the Database Design process.	7M
	(b).	Define these terms: Entity, Entity set, Attribute, Key.	7M

UNIT-II

3.	(a).	What is NULL? What is its importance? How are these values handled in relational model?	7M
	(b).	Consider the following database schema to write queries in SQL Sailor(sid, sname, age, rating) Boats(bid, bname, bcolor) Reserves(sid, bid, day) i) Find the sailors who have reserved a red boat ii) Find the names of the sailors who have reserved at least two boats iii) Find the colors of the boats reserved by ‘Mohan’.	7M

(OR)

4.	(a).	Explain following in brief i. Triggers ii. Assertions	7M
	(b).	Define the terms: Entity Set, Role, Relationship set, Aggregation.	7M

UNIT-III

5.	(a).	How to compute closure of set of functional dependency? Explain with a suitable example schema	7M
	(b).	What is multi valued dependency? State and explain fourth normal form based on this concept	7M

(OR)

6.	(a).	What is the need for schema refinement? Explain with examples.	7M
	(b).	Describe the properties of decomposition	7M

UNIT-IV

7.	(a).	Define transaction and explain desirable properties of transactions.	7M
	(b).	What is a trigger? How to create it? Discuss various types of triggers.	7M

(OR)

8.	(a).	Explain the transaction support in SQL	7M

	(b).	Does two phase locking protocol ensure serializability? Justify your answer	7M

UNIT-V

9.	(a).	Discuss the concept of cylinder and its benefit in disk organization	7M
	(b).	Is B+ tree, a multi level indexing? How does it differ from B-tree?	7M

(OR)

10.	(a).	Explain page formats with examples.	7M
	(b).	Discuss bulk loading of B+ tree with illustrations	7M

[B17CS3213]

[B17CE3206]

III B. Tech II Semester (R17) Regular Examinations

ALTERNATIVE ENERGY SOURCES

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

UNIT-I

1.	(a).	Explain the need for development of renewable energy sources.	7M
	(b).	Explain in detail the scarcity of conventional energy sources in global scenario.	7M
		(OR)	
2.	(a).	Explain hybrid systems in detail.	7M
	(b).	Explain the reduction potential of carbon dioxide gas.	7M
		UNIT-II	
3.	(a).	Explain solar heating and cooling processes with a neat sketch.	7M
	(b).	Explain the measurement and estimation of solar radiation.	7M
		(OR)	
4.	(a).	Explain solar photovoltaic conversion processes.	7M
	(b).	Explain the applications of solar energy conversion processes.	7M
		UNIT-III	
5.	(a).	Derive the expression for power generation from wind turbine using Betz model.	7M
	(b).	List and explain various types of winds and factors influencing wind generation.	7M
		(OR)	
6.	(a).	Explain the design aspects of wind mill.	7M
	(b).	Explain the concepts of energy wheeling and energy banking concepts.	7M
		UNIT-IV	
7.	(a).	Define biomass. Explain how biomass is converted by gasification process.	7M
	(b).	Explain in detail the anaerobic digestion of biomass.	7M
		(OR)	
8.	(a).	Explain how urban waste is collected and converted into energy.	7M
	(b).	Explain how biomass is converted by pyrolysis and liquefaction processes.	7M
		UNIT-V	
9.	(a).	Discuss the theory and working principle of ocean thermal energy conversion (OTEC) system.	7M
	(b).	Explain with a neat sketch, the operation of geothermal power plant.	7M
		(OR)	
10.	(a).	Explain with a neat sketch, the working and operation of any turbine used for hydropower generation.	7M
	(b).	Explain the tidal energy conversion process with a neat sketch.	7M

[B17CE3206]

[B17CE3207]

III B. Tech II Semester (R17) Regular Examinations

WASTE WATER MANAGEMENT

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

UNIT-I

1.	(a).	Explain the physical and chemical characteristics of industrial wastes.	7M
	(b).	Explain the environmental impacts due to improper disposal of industrial wastes.	7M
		(OR)	
2.	(a).	List the various sources of water required for industries. Explain the quality of water required for industrial processing.	7M
	(b).	What is meant by effluent standards? Explain the standards as per BIS for safe disposal.	7M
		UNIT-II	
3.	(a).	Explain the process of heavy metal removal employed in industries.	7M
	(b).	Explain reed bed technology in detail	7M
		(OR)	
4.	(a).	Explain aerobic and anaerobic biological treatment of waste.	7M
	(b).	Explain about equalization and neutralization processes in detail	7M
		UNIT-III	
5.	(a).	Explain the waste reduction by ion exchange process	7M
	(b).	List various membrane technologies used for waste minimization. Explain any one of them in detail.	7M
		(OR)	
6.	(a).	Explain the ozonation technique for waste minimization.	7M
	(b).	Explain the various characteristics of industrial waste water effluent.	7M
		UNIT-IV	
7.	(a).	Explain how sludge is digested by aerobic and anaerobic digesters.	7M
	(b).	Explain different methods of industrial sludge disposal.	7M
		(OR)	
8.	(a).	Explain the effluent disposal methods in detail. Enumerate the advantages and disadvantages of the methods employed.	7M
	(b).	Explain the characteristics of industrial effluent and sludge. What steps are necessary for their safe treatment and disposal.	7M
		UNIT-V	
9.	(a).	Explain the treatment process adopted in tannery industries.	7M
	(b).	Explain the treatment process used in textile industries with a neat sketch.	7M
		(OR)	

10. (a). Explain briefly the characteristics and treatment of cane sugar mill effluent with the aid of a flow chart 7M

(b). Write briefly about the regulatory liabilities in industrial waste water treatment 7M

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[B17CE3208]

III B. Tech II Semester (R17) Regular Examinations

GREEN FUEL TECHNOLOGIES

CIVIL ENGINEERING

MODEL QUESTION PAPER

TIME: 3Hrs. **Max. Marks: 70**

Answer **ONE Question** from **EACH UNIT.**

All questions carry equal marks.

UNIT-I

1.	(a).	Explain why green technology is very important now a days.	7M
	(b).	Explain in detail the factors affecting green technologies.	7M
		(OR)	
2.	(a).	Define green fuel and discuss classification of green fuels.	7M
	(b).	Explain briefly Thermo chemical conversion of biomass to liquids & gaseous fuels.	7M
		UNIT-II	
3.	(a).	Explain about Sugar Fermentation Process.	7M
	(b).	Define Lignocellulosic biomass& explain about its Substrates.	7M
		(OR)	
4.	(a).	Discuss briefly how bioethanol is produced from crops.	7M
	(b).	Explain about Banana Pseudo stem.	7M
		UNIT-III	
5.	(a).	Explain biodiesel production from Algae	7M
	(b).	Is algae culture economical for biodiesel production or not? Explain in detail	7M
		(OR)	
6.	(a).	Define super critical fluid technologies. Classify its types.	7M
	(b).	Explain biodiesel production from super critical fluid technologies.	7M
		UNIT-IV	
7.	(a).	Explain the chemical process of biodiesel production from different plant seeds.	7M
	(b).	Discuss briefly about Palm oil diesel production.	7M
		(OR)	
8.	(a).	How is biodiesel produced from rubber seed oil?	7M
	(b).	What are Vegetable oils? Explain their role in production of biodiesel.	7M
		UNIT-V	
9.	(a).	Explain biogas production by biogas plant.	7M
	(b).	Discuss about biogas technology in India.	7M
		(OR)	
10.	(a).	Explain Microbial production of Methane.	7M
	(b).	Compare and contrast the green fuels with conventional fossil fuels with reference to both environmental and ecological aspects.	7M

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